

Article

Optimizing Sustainability Offshore Hybrid Tidal-Wind Energy Storage Systems for an Off-Grid Coastal City in South Africa

Ladislav Mutunda Kangaji ^{1,*} , Atanda Raji ¹  and Efe Orumwense ² 

¹ Department of Electrical and Electronic Engineering, Cape Peninsula University of Cape Town, Cape Town 7535, South Africa; rajia@cput.ac.za

² Department of Mechanical Engineering, Cape Peninsula University of Cape Town, Cape Town 7535, South Africa; orumwensee@cput.ac.za

* Correspondence: laskangaji@gmail.com

Abstract: South Africa's extensive marine energy resources present a unique opportunity for advancing sustainable energy solutions. This study focuses on developing a sustainable hybrid power generation system that combines offshore wind and tidal current energy to provide a stable, renewable energy supply for off-grid coastal communities. By addressing the challenges of intermittency and unpredictability in renewable energy sources, the proposed system integrates wind and tidal energy with energy storage and diesel backup to ensure reliability while reducing greenhouse gas emissions and minimizing the environmental footprint. The system is optimized for sustainability, with a configuration of one wind turbine, five tidal turbines, and a diesel generator demonstrated to be the most effective in increasing the renewable energy fraction and lowering the net present cost. Simulations conducted using HOMER Pro version 3.20 software underscore the potential of this hybrid system to support South Africa's transition to a more sustainable energy future, aligning with national and global sustainability goals. The results emphasize the environmental benefits of combining these renewable energy sources, offering a blueprint for achieving energy security and sustainable development in coastal regions.

Keywords: green energy solutions; HOMER Pro optimization; hybrid offshore wind and tidal systems; off-grid coastal sustainability; reducing greenhouse gas emissions; renewable energy storage; sustainability in renewable energy



Citation: Kangaji, L.M.; Raji, A.; Orumwense, E. Optimizing Sustainability Offshore Hybrid Tidal-Wind Energy Storage Systems for an Off-Grid Coastal City in South Africa. *Sustainability* **2024**, *16*, 9139. <https://doi.org/10.3390/su16219139>

Academic Editors: Wei Liu and Ming Cheng

Received: 17 August 2024
Revised: 30 September 2024
Accepted: 9 October 2024
Published: 22 October 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Access to electricity is fundamental for modern living, supporting essential services such as healthcare, education, transportation, and communication. Despite this, as of 2016, approximately 1.1 billion people, or 14% of the global population, lacked access to electricity, with rural areas being disproportionately affected [1]. In countries such as South Africa, the disparity in electrification between rural and urban areas is significant, primarily due to the distance from main-grid substations and the substantial costs associated with infrastructure development [2]. While renewable energy sources such as wind, solar, and marine resources present viable solutions, their integration into remote and off-grid systems remains a challenge. Marine energy, encompassing tidal and wave power, offers considerable potential due to its vast and predictable energy reserves [3].

Despite the significant potential of sea and ocean energy resources, these technologies have not yet been widely adopted. Their extensive application is hindered by inherent challenges, including high capital costs and the intermittent and unpredictable nature of energy generation. However, research on wind turbines has surged with recent technological advancements, and the technology employed in wind turbines is now being applied to marine current energy as well [4,5]. Designing such a system must adhere to several constraints. These constraints are divided into two categories: technical, which

includes energy source availability; generation components, battery system state of charge, and electric load location; and economical, which encompasses reliability, flexibility, efficiency, and costs [6]. Energy storage systems offer numerous benefits to electrical energy systems, including improved power quality, mitigation of issues related to intermittent renewable energy sources (RES), balancing generation and demand, and regulating voltage and frequency. Additionally, hybrid energy storage systems (HESS) have been developed by integrating two or more energy storage technologies. This combination leverages the unique characteristics of each technology to achieve the desired performance [7].

For instance, hybrid systems integrating wind, diesel, and batteries are effective in regions with consistent wind resources, while photovoltaic (PV)-diesel-battery configurations are ideal for areas with abundant solar irradiance [8]. Incorporating ESS into hybrid systems can significantly improve the performance of both stand-alone and grid-connected configurations by ensuring a reliable power supply and reducing reliance on fossil fuels.

Although numerous studies have optimized the sizing of ESS for hybrid renewable systems, few have focused on integrating marine energy into microgrids in remote coastal regions. This paper aims to address this gap by evaluating the technical and economic feasibility of a hybrid renewable energy system that integrates tidal and wind energy, supported by ESS, for an off-grid coastal community in South Africa. Optimization Homer Pro version 3.20 software is employed to simulate the system's performance based on hourly load profiles, with the goal of minimizing diesel consumption and reducing electricity costs.

By integrating tidal energy into the microgrid, this research proposes a novel approach to improving system efficiency and stability. This approach is particularly relevant for island microgrids dependent on diesel generators, as it seeks to reduce fossil fuel consumption and enhance power quality. The findings of this study can provide valuable insights for the design of cost-effective and reliable hybrid energy systems in similar contexts, contributing to the broader objective of achieving sustainable energy access.

2. State-of-the-Art Review on Techno-Economic Analysis of Hybrid Energy Systems

The optimal design of hybrid renewable energy systems is a critical area of study, with an extensive body of literature dedicated to this topic. The core design challenge involves determining the optimal configuration of power systems, including the ideal location, type, and sizing of generation units at specific nodes, to meet load requirements at the lowest possible cost [8].

Much of the existing research focuses on the application of algorithms for the optimal sizing and economic evaluation of renewable energy systems, particularly those involving photovoltaics (PV), wind, and micro-turbine-based HRES, with battery storage elements connected to the utility grid. For example, Liu et al. [9] proposed a frequency-based method for sizing a battery-super capacitor (SC) hybrid energy storage system (HESS) to maintain power balance in an isolated system with significant wind penetration, utilizing Fourier transformation. Meanwhile, Jacob et al. [10] proposed a pinch analysis-based strategy for sizing hybrid energy storage systems (HESS) in photovoltaic-based isolated power systems, where generation consistently exceeds load. Arabali et al. [11] introduced a search-based optimization approach to determine the optimal size of energy storage systems and to analyze the reliability of hybrid wind-solar power systems, employing probability distribution functions to account for the stochastic nature of renewable energy sources (RES) generation.

To smooth the output power from a wave energy park, the output power from a wave energy park Said et al. [12] utilized batteries and SC modules, assessing the power delivered to the grid as the average output of the wave energy conversion system (WECS) over a specified time. They sized the HESS based on the power balance between the wind energy conversion system (WECS) generation and the power delivered to the grid, considering both power and energy capacity. However, the detailed sizing of individual battery and supercapacitor (SC) components was not thoroughly examined. Lafoz et al. [13] developed a program for sizing HESS to power onboard loads using autonomous WECS. The program

calculated the probability and amplitude of each component in the WECS output power spectrum based on the known location of the energy harvesting devices. The generated output power profile was then used to size the HESS over short, medium, and long periods, considering its probability of occurrence. To avoid oversizing, the HESS sizing was derived from a long-term operation analysis, balancing load-energy demand [14].

This research aims to determine the optimal size for an offshore standalone tidal-wind-battery storage (BS) hybrid system that is more cost-effective. Specifically, it seeks to find the optimal size for each component of the hybrid renewable energy system (HRES) to allow consumers to recover their investment over time by reducing the cost of electricity purchased from the grid. Given that solar and wind energy are location-dependent, this study is based on the meteorological data of West Bank, East London, a coastal city in South Africa. This strategy is applied to an off-grid offshore hybrid tidal wind storage energy system to demonstrate its effectiveness, with potential applicability to other renewable energy configurations.

3. Mathematical Modelling of Offshore Hybrid Power Generation Systems

3.1. Hybrid Wind-Tidal Arrays and Power Generation

The (HRES) are a mixture of many renewable energy sources. It can be connected to the grid or to be used as a stand-alone. The most important benefit of HRES is the optimum use of sustainable energy generation working characteristics, which results in efficiencies higher than that could be reached from one power source of HRES designed for the specific location [15]. When designing a hybrid (HRES), many aspects need to be considered, such as reliability and cost, which are the key aspects of the implementation; according to many studies, hybrid energy systems are more reliable and cost-effective than systems based on just a single source [1,8,16,17].

One of the primary advantages of a hybrid energy system is its ability to bridge energy shortfalls from one source by augmenting with another source, particularly during weather changes [18–20]. For instance, when sunlight is insufficient to power photovoltaic panels or when wind speeds are inadequate for turbine generation, the hybrid system can compensate by leveraging the alternative energy source to maintain consistent power generation [21].

3.2. Wind Power Generation

Wind is the motion of air created by the irregular heat of the earth's surface. It has been happening at all measures, from local breezes made by heating of surfaces of the land that remain some temps to global winds created from the heating of solar at the earth [15]. The practical application of wind energy is primarily the conversion of wind energy into other forms of energy; a process usually achieved through wind turbines, which capture and convert the power of wind currents.

3.2.1. Wind Turbines Component Description

Figure 1 illustrates the components of a wind power system, including a wind turbine, battery, DC-DC converter, DC-AC converter, AC-DC converter, and loads, which can be configured with either a horizontal or vertical axis.

Wind power is harnessed through wind turbines, which capture the kinetic energy of the wind. This energy is converted into mechanical energy transferred through the drive train to the generator, which converts it into electrical energy [22]. Subsequently, the rotational energy is transformed into electrical power through the transmission system.

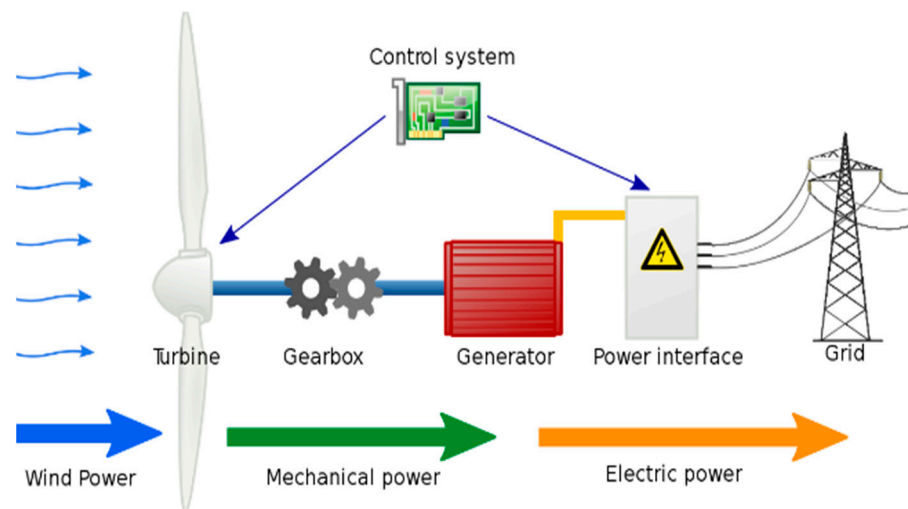


Figure 1. Block diagram of wind turbine.

3.2.2. Modelling of Offshore Wind Energy System

A significant supply of renewable energy will come from offshore wind. Onshore wind speed variation is less of a range than offshore wind speed variation [23], where ρ is the density of air in 1.22 kg/m^3 . The captured mechanical power P_W , the dimensionless power coefficient C_{PW} , the measured mechanical torque T_W , the tip speed ratio λ , and the blade pitch angle β [24]. The tip speed ratio changes when the rotor speed and wind speed change, resulting in a new operating point on the power curves at various speeds and pitch angles [25]. At rated wind speed, the generating power equals the turbine's nominal power. If the wind speed grows, the output power may increase until the nominal turbine power is reached. Wind speeds may continue to rise, so there is a risk of damaging the rotor of the turbine [26].

The kinetic energy of a wind turbine rotor with the cross-sectional surface (A) exposed to the tidal stream is expressed as:

$$E_W = \frac{1}{2} \rho v V^2 \quad (1)$$

where v represents the parcel volume of air interacting with the rotor, and ρ denotes the water density. The interaction between the air parcel and the rotor occurs over a cross-sectional area A , corresponding to the rotor's swept area, with a thickness determined by the air velocity V . The generator efficiency is represented by η_g [27]. Thus, the power, which is the kinetic energy per unit time, is expressed as:

$$P_{WE} = \frac{1}{2} \rho A V^3 C_P(\lambda, \beta) \eta_t \eta_g \quad (2)$$

Figure 2 illustrates the corresponding power curve of the wind generator at its rated power output P_{WE} [28]. The power coefficient of a turbine depends on both the Tip Speed Ratio (TSR) and the blade pitch angle β .

For a given turbine, the power coefficient depends on the TSR and the blade pitch angle. The typical variation of the power coefficient concerning the TSR for various values of the pitch angle [28].

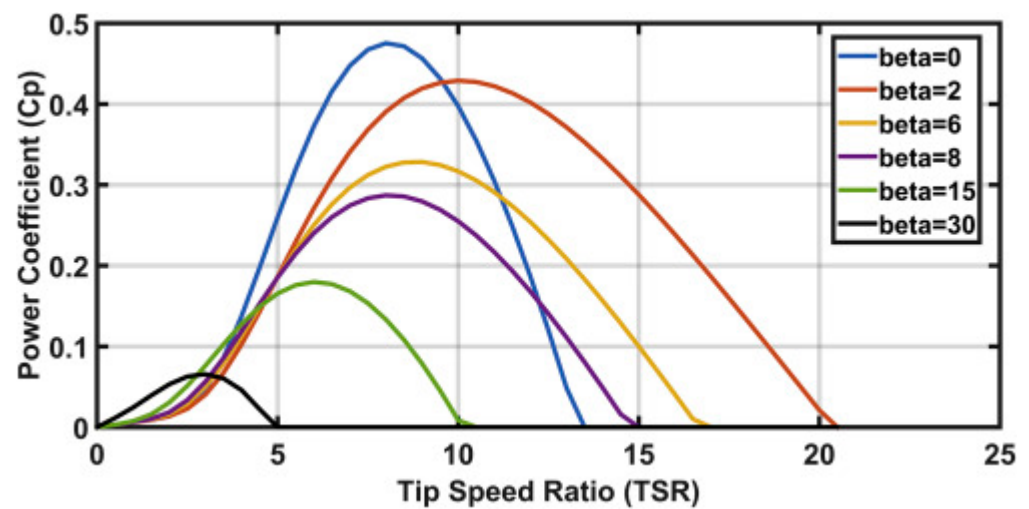


Figure 2. C_p and λ (TSR) for various pitch angles β [29].

3.2.3. Wind Turbines Power Curves

Wind turbine is used. Some of its parameters are described in Table 1.

Table 1. Wind turbine parameters.

Parameters	Values
Cut-in speed, v_{cutin}	5 m/s
Cut-out speed, v_{cutout}	25 m/s
Rated speed, v	15 m/s
The maximum output power, P_{wmax}	2300 W
Swept area	3959 m ²
Number of blades	3
Rotor diameter	32 m, 40 m, 50 m
Hub height	32 m, 5 m

The height of the wind turbine tower is a critical factor that significantly influences the turbine's operating performance and can account for more than half of the total system cost. To adjust the measured wind speed to the hub height, Equation (3) is used.

$$v(t) = v_r(t) \left(\frac{h}{h_r} \right)^\gamma \quad (3)$$

where $v(t)$ represents the wind speed at the desired height h , $v_r(t)$ is the wind speed measured at a known reference height h_r , and the power law exponent.

γ is a factor that depends on the roughness of the terrain [30]. For this study, the power law exponent γ is set to 0.2 [31]. The second figure in Section 5.5 shows the wind speeds on the considered island and the corresponding wind turbine power production at various hub heights [32].

3.3. Tidal Energy Generation Systems

The technology used for extracting wind energy is similar to that used for harnessing tidal energy, as indicated in Figure 3. As a result, many wind generator models can also be applied to tidal energy systems [33]. Figure 3 illustrates a turbine configuration based on a Permanent Magnet Synchronous Generator (PMSG) paired with a Full Power Converter (FPC), which is suitable for both Tidal Stream Turbines (TSTs) and Wind Turbines (WTs) [34]. This setup allows for variable-speed operation since the electrical frequency of the PMSG can vary with changes in wind or tidal flow speed while the grid frequency remains stable.

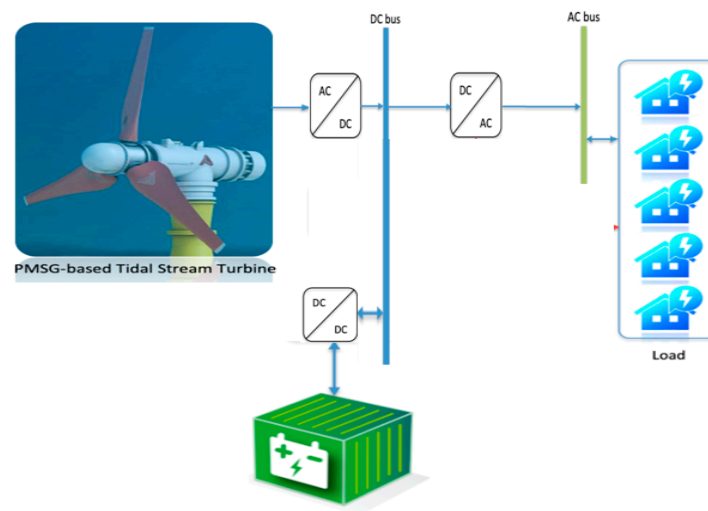


Figure 3. Tidal energy systems configuration based on a PMSG.

Tidal Energy Modelling

The objective of the Tidal Stream Turbine (TST) or Wind Turbine (WT) is to maximize power extraction from the tidal stream or wind flow. The power transferred to the turbine rotor and the resulting torque are determined by the following equations:

$$P_t = C_P P_{flow} = \frac{1}{2} A V_{flow}^3 C_P(\beta, \lambda) \quad (4)$$

$$T_t = \frac{P_t}{\omega_{rot}} \quad (5)$$

where P_{flow} (W) represents the power of the tidal stream flow; A (m^2) is the swept area of the rotor; V_{flow} (m/s) is the upwind tidal stream flow speed; C_P the power coefficient; and ω_{rot} (rad/s) is the rotor angular speed [16].

In Equations (4) and (5), C_P represents the turbine's performance and efficiency as a function of the blade pitch angle and the tip-speed ratio (TSR) [35] where R (m) is the rotor's radius at the tip. At a single TSR (at a certain wind or flow speed) and at a fixed rotational speed of the turbine, a maximum C_P is attained. Thus, a maximum C_P can be attained over a range of wind/flow speeds, provided the turbine is run at a variable speed [36]. Figure 4 illustrates a typical feature with the optimal power extraction-speed curve crossing C_{Pmax} spots for various wind flow speeds. Table 2 provides the layout for the tidal Parameters and cut-out tidal parameters [1].

Table 2. Tidal parameters.

Parameters	Symbol	Values
Cut-in tidal speed	v_{cuttid}	< 1 (m/s)
Rated tidal speed	$v_{ratedtid}$	3 m/s
The cross-sectional area of the turbine	A	201.06 m^2
Power coefficient	C_p	0.44
Cut-out tidal speed	$v_{cututtid}$	> 5 m/s

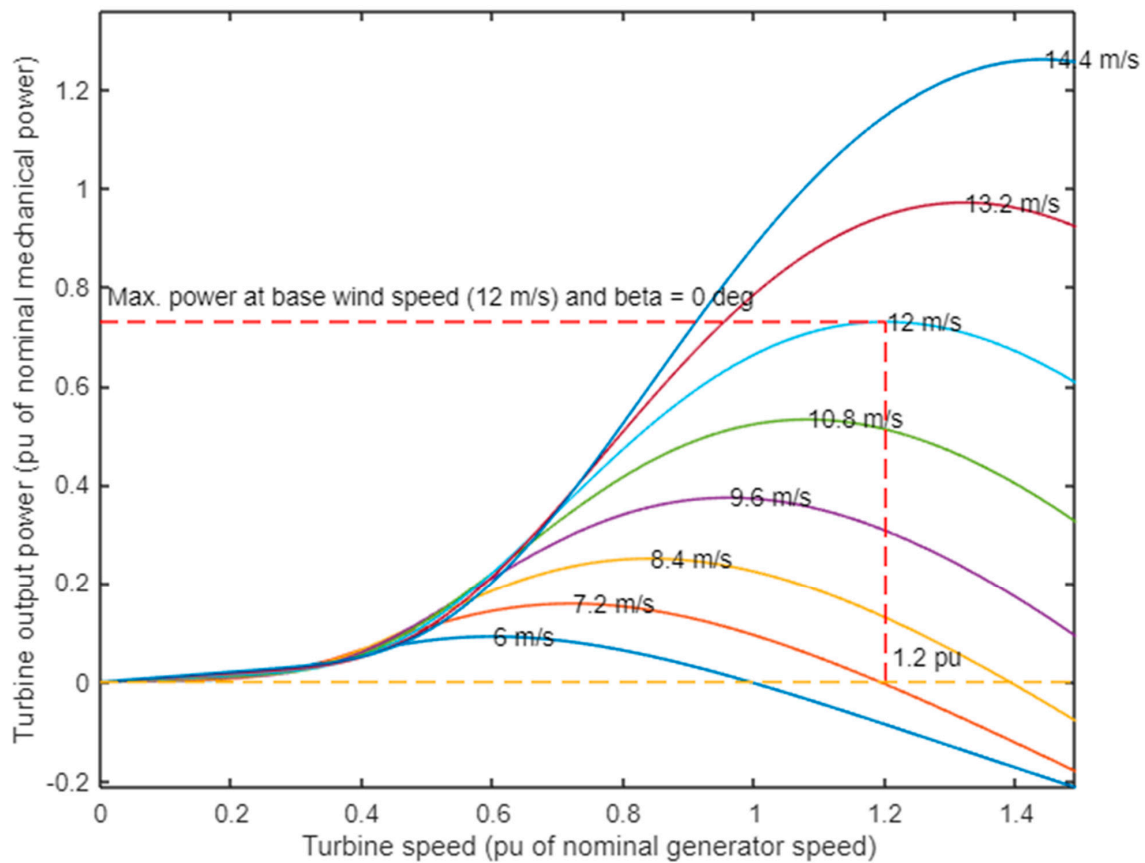


Figure 4. Characteristic for maximum power extraction.

Additionally, the generator torque-speed characteristic is displayed. As the wind/flow speed changes, the control goal is to keep the turbine on this curve [37,38].

Power is restricted to prevent overloading when the cut-out speed reaches the rated torque. To decrease the lift force in the turbine's blades, if the blades have variable pitch, they are pitched to feather. In this manner, power is controlled and the torque on the rotor is decreased. In contrast, if the turbine's blades have a set pitch, they can be adjusted such that the angle of attack rises, so purposefully inducing stall [39].

3.4. Battery Bank

The fluctuations in wind energy generation often do not align with the timing of energy demand. Consequently, power generation systems require the integration of battery storage facilities to address the temporal mismatch between load requirements and the availability of wind and tidal energy, as well as to ensure system maintenance. In small-scale, stand-alone power systems, batteries represent a significant cost factor. Battery stacks can consist of multiple units, ranging from 0 to 500. For this study, a 6 V, 1.34 kWh battery was selected, with each string containing 8 batteries [40].

3.5. Battery Energy Storage System Modelling

The battery's state of charge (SOC) is the cumulative sum of daily charge/discharge cycles. At any given hour (t), the SOC is related to the state of the previous hour and the energy production and consumption within the system during the interval from $t - 1$ to t . During charging, when the total system output exceeds load demand, the available battery capacity at hour (t) can be described by [40]:

$$E_{Bat}(t) = \text{Battery Capacity} \frac{Ah}{= \text{Battery}} \text{Current (A)} \quad (6)$$

Conversely, when load demand surpasses the generated energy, the battery bank discharges, and the available capacity at an hour (t) is given by:

$$E_{Bat}(t) = E_{Bat}(t-1) - E_{need}(t) \quad (7)$$

Let d represent the ratio of the minimum allowable SOC voltage limit to the maximum SOC voltage when the battery is fully charged. The Depth of Discharge (DOD) is then defined as [40]:

$$DOD = (1 - d) * 100 \quad (8)$$

3.6. Converter Model

The rectifier model is given below:

$$E_{REC} - IN(t) = E_{SUR} - AC(t) \quad (9)$$

$$E_{SUR} - AC(t) = E_{DG}(t) \quad (10)$$

$E_{SUR} - AC(t) = E_{DG}(t)$ is the amount of surplus energy from AC source, kWh, $E_{DG}(t)$ is surplus energy from a diesel generator to DC power of constant voltage when the energy generated by the hybrid energy system exceeds the load demand [40].

3.7. Diesel Generator Characteristics and Modelling in HOMER

3.7.1. Diesel Generator Characteristics

In renewable energy hybrid systems, diesel generators serve as backup power sources to ensure reliability and stability. For effective mode ling and optimization in HOMER, the following key characteristics of diesel generators should be considered:

- **Rated Capacity:** This refers to the maximum electrical output the generator can produce, typically measured in kilowatts (kW).
- **Fuel Consumption:** This is the amount of diesel fuel consumed per unit of electricity generated, usually provided in liters per hour (L/h) or gallons per hour (GPH). Fuel consumption is often depicted as a fuel curve, illustrating consumption at different load levels.
- **Efficiency:** This is the ratio of useful electrical output to the total fuel energy input, expressed as a percentage. Efficiency can vary depending on the load.
- **Operational Constraints:** These include the minimum and maximum load levels at which the generator can operate effectively. Operating below the minimum load can result in inefficiencies and increased maintenance requirements.
- **Emissions:** The number of pollutants (CO_2 , NO_x , SO_2 , PM) emitted per unit of fuel consumed. These emissions are important for environmental impact assessments.
- **Maintenance and Lifespan:** Regular maintenance requirements and the expected lifespan of the generator, typically measured in operating hours or years.

3.7.2. Load Model

The number of electrical power sources in a hybrid renewable system is determined by the electrical load demand. The load profile for the off-grid coastal village community under study, with a maximum demand of 20 MW. Peak load typically occurs in June and July.

4. Materials and Methods

Meteorological Data

This study used weather station 00595729, located in the West Bank–East London region of Buffalo City Metropolitan Municipality (BCMM), Eastern Cape Province, South Africa. As shown in Figure 5, the station is situated at latitude -33.0330° S and longitude 27.8330° E and monitors the entire West Bank–East London area (<https://www.google.co>

[za/maps/search/West+bank+East+london+community/@-33.0352403,27.8987226,1390m/data](https://www.google.com/maps/search/West+bank+East+london+community/@-33.0352403,27.8987226,1390m/data), accessed on 10 December 2023). The meteorological data utilized in this research were gathered from coastal villages in the West Bank, South Africa [41]. The dataset comprises average hourly data spanning a full year (8760 h), starting in January 2024. This study assesses the feasibility of an offshore hybrid wind-tidal storage diesel generator energy system using Homer Pro version 3.20.

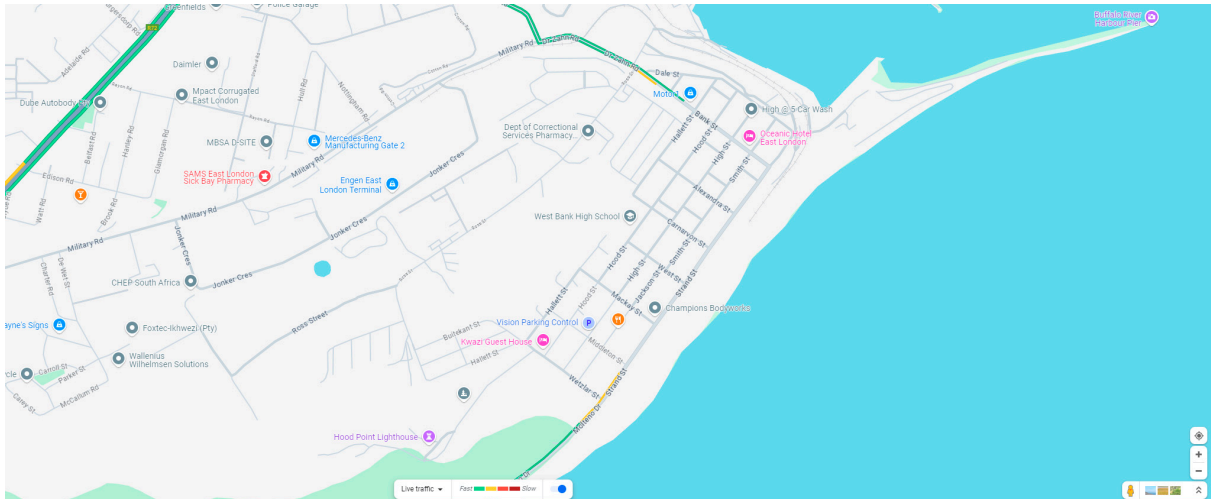


Figure 5. The West Bank-East London area in the Eastern Cape Province of South Africa.

5. Methodology for Optimizing Offshore Hybrid Energy Systems

5.1. Summary of Methodological Steps for the Proposed Study

This diagram illustrates the general model for optimizing a hybrid generation system, which integrates multiple energy sources such as wind, solar, and diesel. It outlines the key components and their interactions, including power generation units, storage systems, and control strategies. The diagram highlights the optimization goals, such as minimizing cost and maximizing efficiency, while balancing energy supply and demand [42]. The connections between components and their roles in the overall system are depicted in Figure 6, providing a comprehensive overview of the hybrid system's functionality and optimization process [6]. This work aims to design an offshore hybrid wind-tidal energy system with a battery bank for a coastal town in South Africa using Homer software. Electrifying remote areas requires a reliable system with minimal maintenance, as frequent repairs and replacements can be challenging.

5.2. HOMER Software

To address the electricity demand of a coastal community, a systematic approach is required, consisting of the following steps as outlined in Figure 7:

- **Site Selection:** Identify a suitable location for the case study.
- **Electricity Demand Assessment:** Gather and analyze load profile data to understand the community's electricity needs.
- **Microgrid Design Definition:** Outline the microgrid components, including multiple energy sources (turbines and generators), storage devices (batteries), an AC to DC rectifier (converter), and a predefined electrical control system (controller).
- **Environmental Data Collection:** Obtain relevant data for energy resources, such as fuel type, wind speeds, and water currents.
- **Design Optimization:** Determine the optimal location, size, and quantity of devices necessary to harness the available energy resources effectively.
- **Feasibility and Financial Analysis:** Evaluate the optimized microgrid design's financial viability and overall feasibility.

To calculate the levelized cost of a hybrid energy system, it is essential to identify and include various individual costs, though estimating costs related to permissions and licenses can be challenging. These costs are often difficult to predict because the regulatory framework is not yet fully established and varies by country. The capital costs, which cover the purchase and installation of system components, must be factored into the optimization process of the hybrid energy system [43]. Equally important is the cost of maintenance over the system's lifetime, especially since these installations are exposed to harsh marine conditions. Decommissioning costs must also be considered. Additionally, to calculate the levelized cost [44]. Finally, the decommissioning cost must be included. In addition, to calculate the levelized cost [45], it is crucial to know the capacity factor of each technology to determine power output, as well as the discount rate and the return on investment required to make the project financially viable [46]. Offshore hybrid energy systems must be both cost-effective and highly reliable to meet load requirements [47]. To design a renewable-based hybrid energy system, it is essential to address a range of constraints, which fall into two key categories: technical and economic. Technical constraints include factors such as the availability of energy sources, the state of charge of battery systems, the performance of generation components, and the location of the electrical load. On the economic side, considerations such as reliability, flexibility, efficiency, and overall costs come into play [48]. To evaluate and optimize these systems, various commercial software tools are available, including HOMER, RET Screen, Hybrid2, TRNSYS, RAPSIM, and INSEL. Among these, HOMER, developed by the National Renewable Energy Laboratory (NREL), is one of the most widely used. It excels at simulating, modeling, and optimizing hybrid power configurations for off-grid systems [49,50].

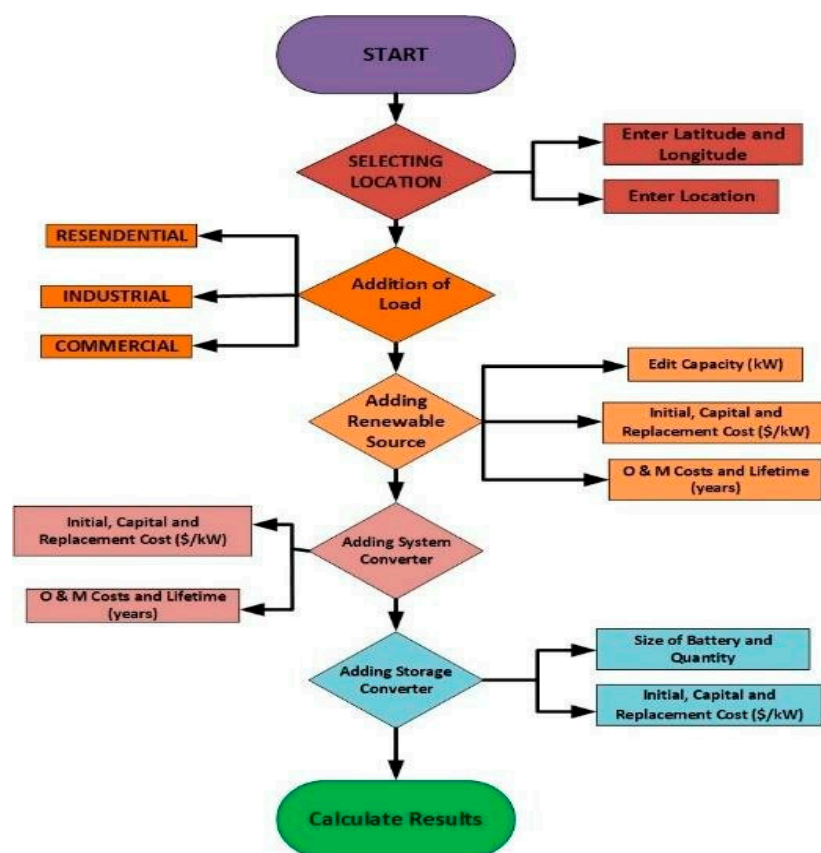


Figure 6. Hybrid generation system optimization general model.

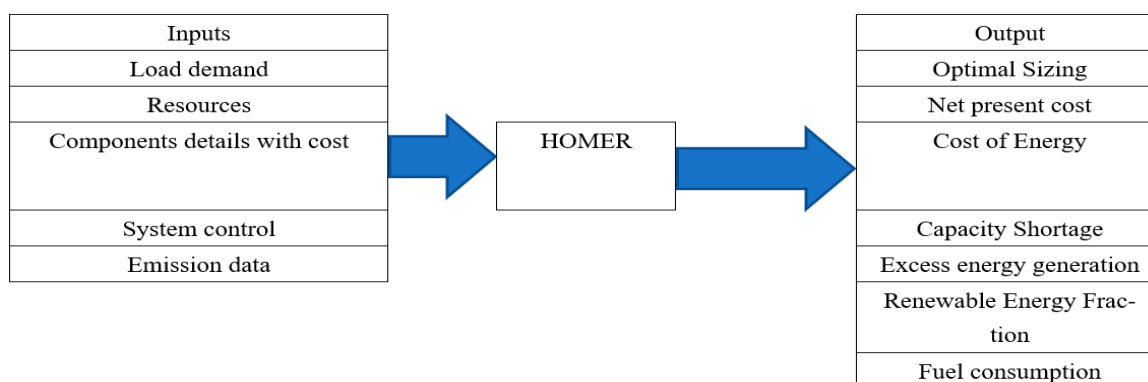


Figure 7. Schematic representation of HOMER.

5.3. Hybrid Tidal/Wind/Battery System

Figure 8 illustrates the platform diagram of the combined wind and tidal power generation system. Both the wind turbine and the tidal current turbine use horizontal shaft turbines, which are known for their high power generation efficiency [51]. The wind power generation device is installed above sea level, while the tidal power generation device is installed below sea level. By sharing the same base, these devices conserve space and significantly reduce both initial investment and ongoing maintenance costs. The primary types of wind power generation employed are doubly-fed induction generators and permanent magnet synchronous generators.

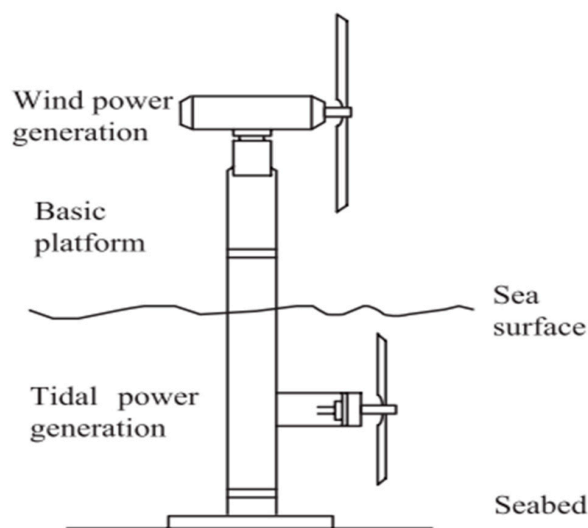


Figure 8. Diagram of the platform for the combined wind and tidal current power generation system.

5.4. Case Study

The research is conducted at a weather station located in the West Bank–East London region of Buffalo City Metropolitan Municipality (BCMM), Eastern Cape Province, South Africa. Positioned at latitude 33.0330° S and longitude 27.8330° E, this station monitors the entire West Bank–East London area. The study area features an ecosystem with ecologically sensitive vegetation. It is bordered geographically by the Indian Ocean to the south, the Igoda River to the west, East London City to the east, and the Buffalo River to the north. The terrain is characterized by rolling coastal plains, rock outcrops, and hills divided by steep valleys [41].

5.5. Loads Profile

The village, which lacks historical electrical load data [1] due to being non-electrified, relies on assumptions to estimate its electrical load:

Peak demand is estimated at 4500 kW for 150 houses in the case study [1], assuming an 80% load diversification for the base load without accounting for area-specific variations. Connected to the larger Bisho customer load network, the village adopts a scaled-down version of the Bisho demand profile (peaking at ~597 kWh) to project its demand. Detailed load data is recorded hourly, covering 8760 h (one year) for input into HOMER Pro[®]. The load primarily consists of residential properties, exhibiting two distinct peaks—smaller in the morning and larger in the evening—as illustrated in Figure 9, depicting average hourly loads throughout the year. Figure 10 presents yearly average loads, with higher expected demand during winter months (June–August) [52].

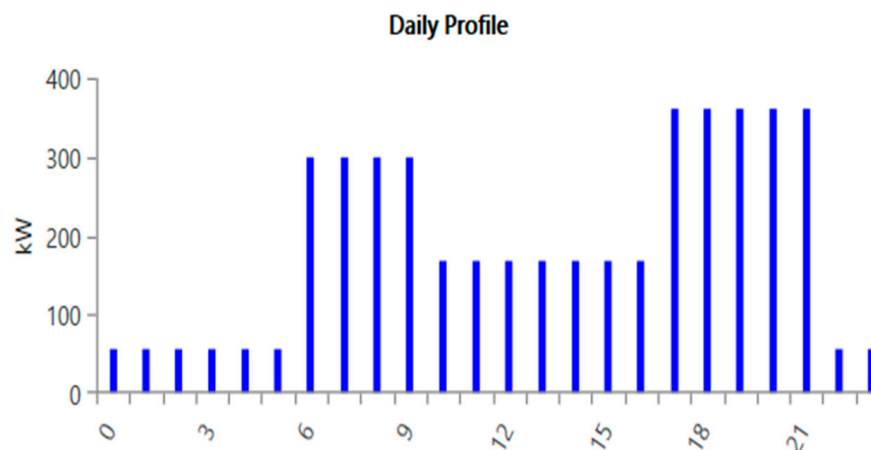


Figure 9. Variation of load demand for 24 h.

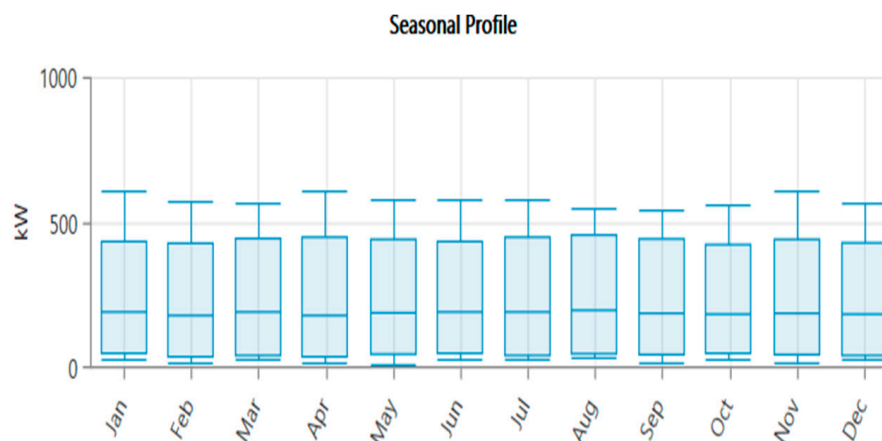


Figure 10. Seasonal profile of load demand throughout the year.

The seasonal load profile in Figure 10 highlights the fluctuations in electricity demand throughout the year for the West Bank community. Typically, there is higher demand during colder months due to increased heating needs and potentially during hot summer months if air conditioning is widely used. This profile underscores the peaks and troughs in electricity consumption, reflecting the community's usage patterns as influenced by seasonal changes.

According to a study conducted by Prinsloo and Davidson, the African village community, a house consumption is around 12.5 kW operating from 9:00 to 11:00 and 15:00 to 18:00. The bulbs can be lit from 18:00 to 23:00. A radio consumption (5 W) and a TV set (70 W) are to be switched usually around 6:00 p.m. to 11:00 p.m. [53].

To distribute the 20 MW load demand throughout the day, we'll consider typical consumption patterns for residential, commercial, industrial, and public service sectors. Here's a detailed breakdown of the daily load profile based on a total load demand of

20 MW; identifying Demand for West Bank, East London, South Africa Village Supply Authority data has been utilized in this case study to estimate monthly and hourly generation. The data indicates that peak values occur in June (69.29 kW) and during the 6–7 p.m. time slot (69.29 kW), while the lowest values are observed in January (162 kW) and between 3 and 4 p.m. (37 kW).

Tidal current data have been sourced from literature review papers [53]. For wind data in South Africa, the Global Wind Atlas extrapolates information from the National Renewable Energy Laboratory and NASA’s Prediction of Worldwide Energy Resources:

- Residential Loads (40% of total demand):
- Morning Peak (6 a.m.–9 a.m.): 25% of residential consumption
- Midday Dip (9 a.m.–4 p.m.): 15% of residential consumption
- Evening Peak (5 p.m.–10 p.m.): 40% of residential consumption
- Nighttime Baseline (10 p.m.–6 a.m.): 20% of residential consumption
- Commercial Loads (30% of total demand):
- Business Hours (8 a.m.–6 p.m.): 60% of commercial consumption
- Off-Peak Hours (6 p.m.–8 a.m.): 40% of commercial consumption
- Public and Community Services (10% of total demand):
- School and Public Buildings (7 a.m.–4 p.m.): 70% of public services consumption
- Evening Community Activities (4 p.m.–10 p.m.): 30% of public services consumption.

Key Considerations

- Peak Demand Management: Understanding when peaks occur helps in planning storage and backup systems to ensure reliability during high-demand periods.
- Renewable Integration: Match the generation profile of wind and tidal resources to the community load profile to maximize renewable energy utilization.
- Energy Efficiency Measures: Encourage energy-saving practices during peak hours to reduce overall demand and improve system stability.
- Scalability: Design the system to be scalable to accommodate future growth in the community’s energy needs.

Table 3 gives a rural African village’s hourly power energy load profile suitable for micro-CHP computer-modulated and simulated experimentally. According to these researchers, an electric load of a house in the rural villages of South Africa can be assumed to be composed of light bulbs, radio, television, 2 plate stove, iron, kettle, fan, and refrigerator t [53]. The load combination and energy demand estimation in Table 3 have been selected.

The environmental parameters of the site are comprehensively detailed in Table 4:

Table 3. The consumption of commonly used electrical appliances.

Appliances	Energy Consumption (Watthours)
Light bulbs	60 W × 5 h
Radio	50 W × 6 h
2Plate stove	2000 W × 2 h
Iron	1000 W × 1/4 h
Kettle	2000 W × 1/4 h
Refrigerator	5 W × 18 h
Fan	6 W × 7 h
Television(colour)	50 W × 6 h

Table 4. Key location parameters for tidal and wind energy assessment.

Location	Latitude (deg)	Longitude (deg)	Annual Average Water Speed (m/s)	Water Depth (m)	Annual Average Wind Speed (m/s)
West Bk-EL	−33.0330 S	27.8330 E	0.52	30	8.31

Current data on Tidal were obtained from the official SAN Hydrographic Office website. The simulation included high-resolution nested domains, specifically targeting the region around East London Harbour. Figure 11 illustrates the bathymetry and the environmental parameters of the site indicated in Table 4 around southern Africa, highlighting the major coastal towns such as Port Nolloth (PN), Saldanha Bay (SB), Cape Town (CT), Mossel Bay (MB), Port Elizabeth (PE), East London (EL), Durban (DN), Richards Bay (RB), and Luderitz [54,55]. Thus, the annual water speed for West Bank East London South Africa is presented in Figure 12.

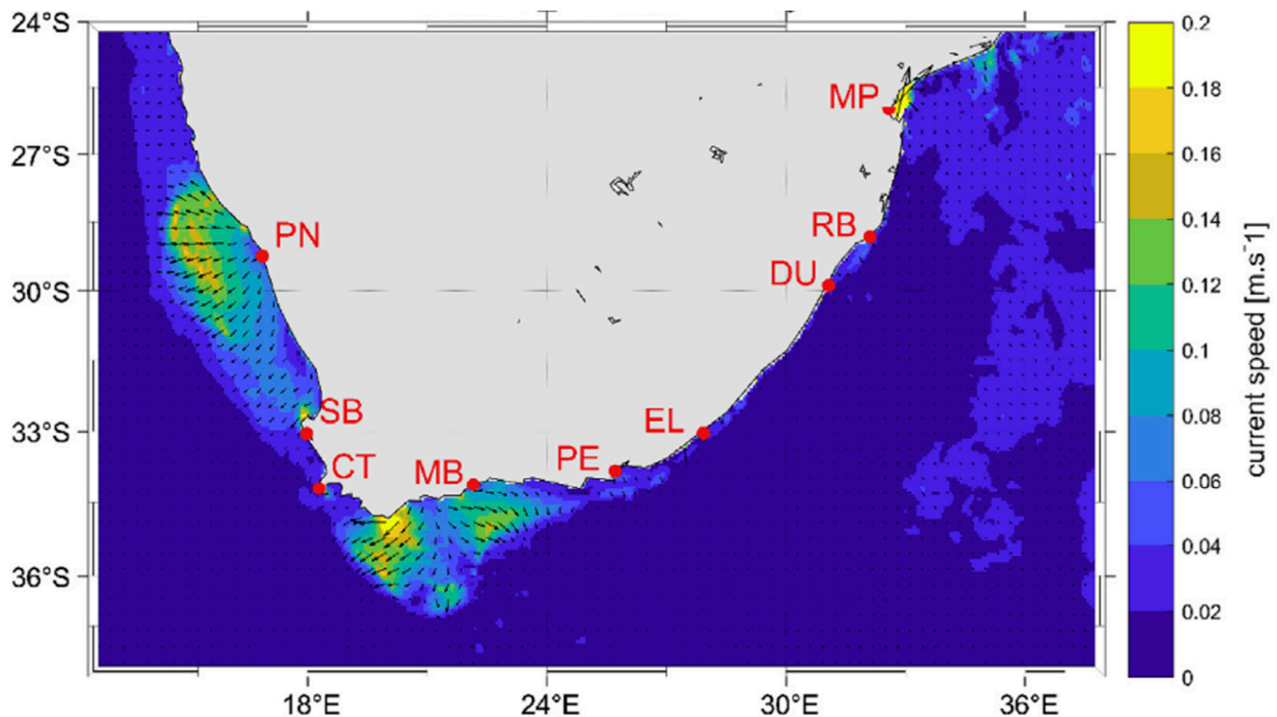


Figure 11. Bathymetry and key coastal cities of Southern Africa [54].

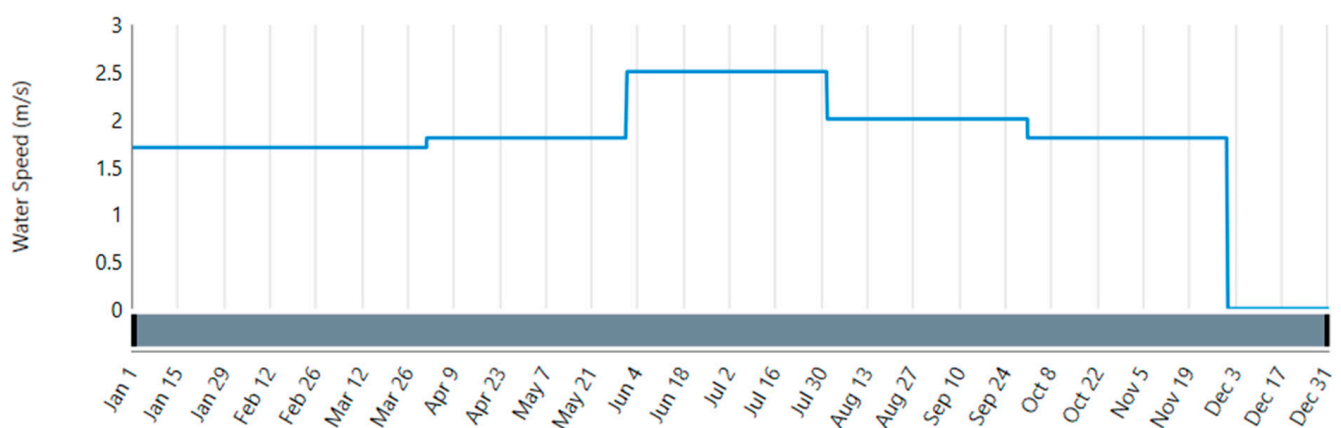


Figure 12. Annual water speed for West Bank East London South Africa.

In the West Bank area of East London, South Africa, wind resources indicated in Figure 13 (<https://globalwindatlas.info/en/area/South%20Africa>, accessed on 10 December 2023), the annual average wind velocity, and the wind speed histogram (as shown in Figures 14 and 15) illustrate their complementarity. Wind power is more abundant in winter and less so in summer, while tidal current energy is less abundant in winter and more so in summer [56,57]. This seasonal complementarity ensures that the two

energy sources can balance each other. The combined utilization of wind and tidal energy improves the stability of the power generation system’s output.

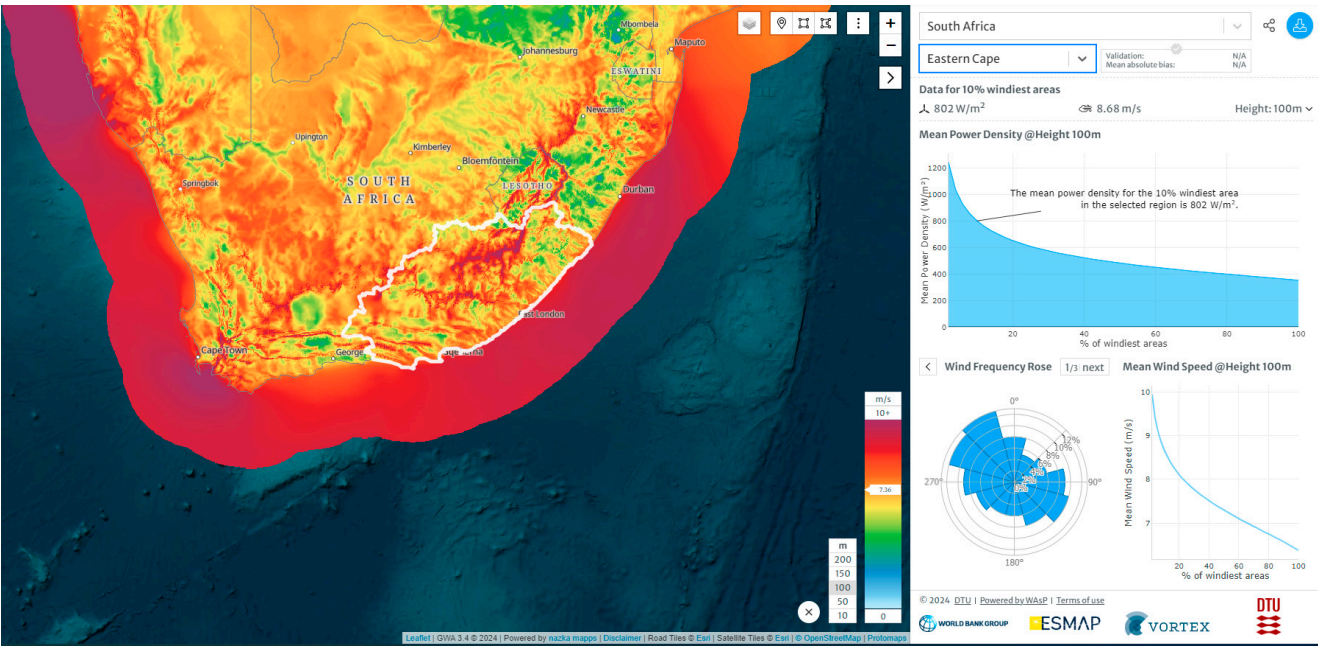


Figure 13. South Africa’s offshore wind resource.

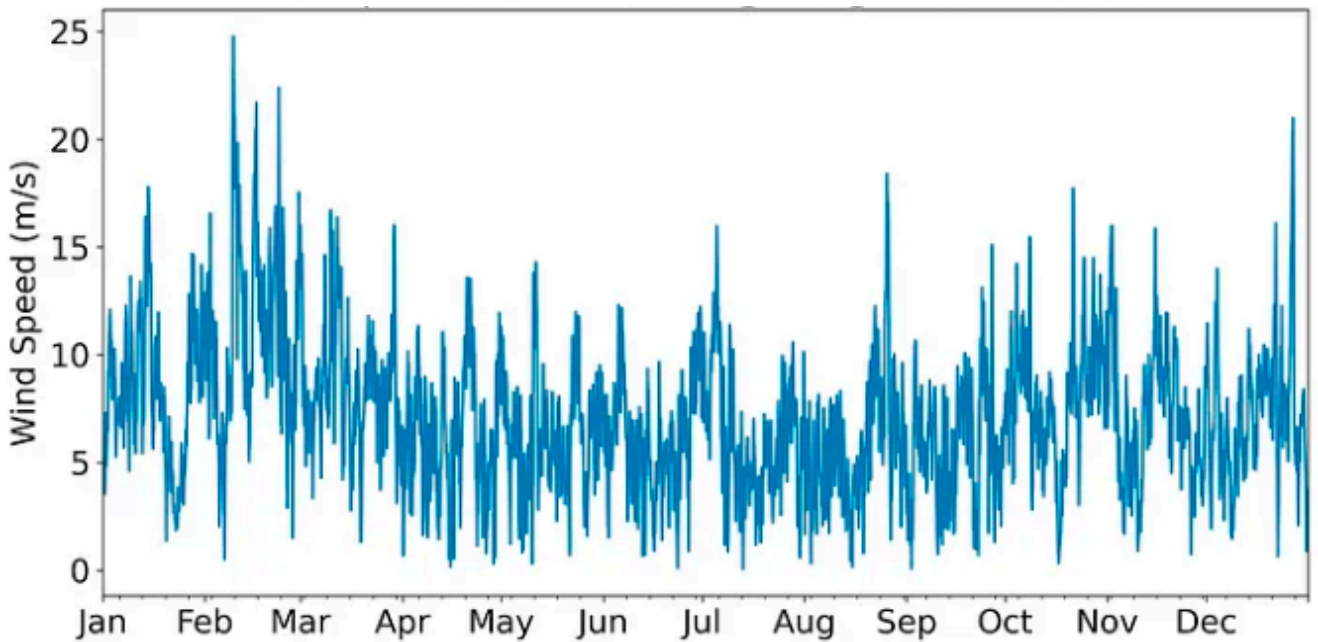


Figure 14. Annual wind speed for West Bank East London South Africa.

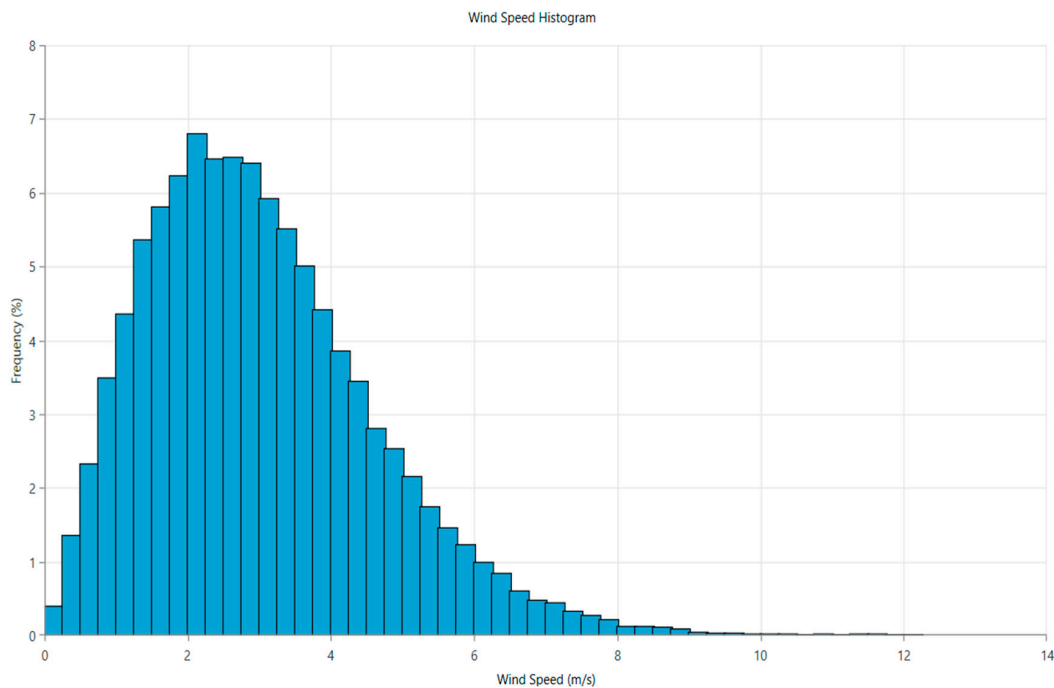


Figure 15. Wind speed histogram.

6. Microgrid Components

This study focuses on offshore microgrid energy systems, both grid-connected and off-grid. The microgrid system under analysis includes wind and tidal turbines and battery energy storage systems (BESS), as illustrated in Figure 16. All components are interconnected via both AC and DC buses. Grid-connected microgrid energy systems offer several advantages, including enhanced reliability, improved power quality, and the ability to sell excess energy back to the main grid. Off-grid microgrid energy systems, on the other hand, provide energy independence, increased resilience in remote or disaster-prone areas, and the potential for cost savings by reducing reliance on traditional energy sources.

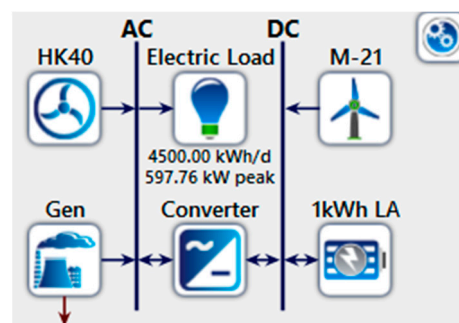


Figure 16. Off-Grid Microgrid Systems: Tidal, Wind, and Storage Integration.

6.1. Wind Turbine

The XANT M-21 (XANT, Brussels, Belgium) 100 kW wind turbine was selected for its low cut-in speed and its capacity to effectively meet peak load demands while providing a substantial share of the peak load at minimal overall costs. With a capital and replacement cost of \$300,000 each and an annual operation and maintenance (O&M) cost of \$10,000, the turbine offers a cost-effective solution over its 20-year lifespan [2]. The turbine's performance, including its power output at 4 m/s wind speed intervals, is shown in Figure 14 and explained in Equation (11).

$$P_{Wind} = 0.0018 V^6 - 0.07V^5 + 1.0826 V^4 - 8.4544 V^3 + 36.229 V^2 - 78.056 V + 66.38 \text{ for } V \leq 10 \text{ m/s} \quad (11)$$

Additional details for the XANT wind turbine are provided in Table 5.

Table 5. Wind System Components.

Wind Turbine	XANT M-21 (100 kW)
Model name	XANT
Rated power	100 kW
Cut-in wind speed	3 m/s
Cut-out wind speed	20 m/s
Hub height	31.8 m
Swept area	346.36 m ²
Rotor diameter	21 m
Capital cost	\$300,000
Replacement cost	\$300,000
O&M	\$10,000
Lifetime	20 years

HOMER Pro software was used to generate wind frequency distribution data for the West Bank-East London area based on downloaded wind speed data from the software. As depicted in Figure 17, wind speeds exceed the XANT M-21 turbine's cut-in speed for 90.5% of the year, confirming that this wind turbine is well-suited for electricity generation at the West Bank-East London site. Similarly, as shown in Figure 18, water speeds exceed the cut-in speed for the Schottel 54 kW tidal turbine for 68.4% of the year, making it an appropriate choice for tidal energy generation at the same location.

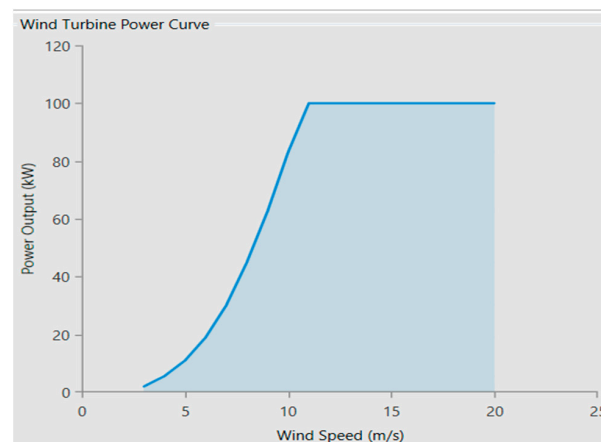


Figure 17. The power curve of the XANT M-21 turbine.

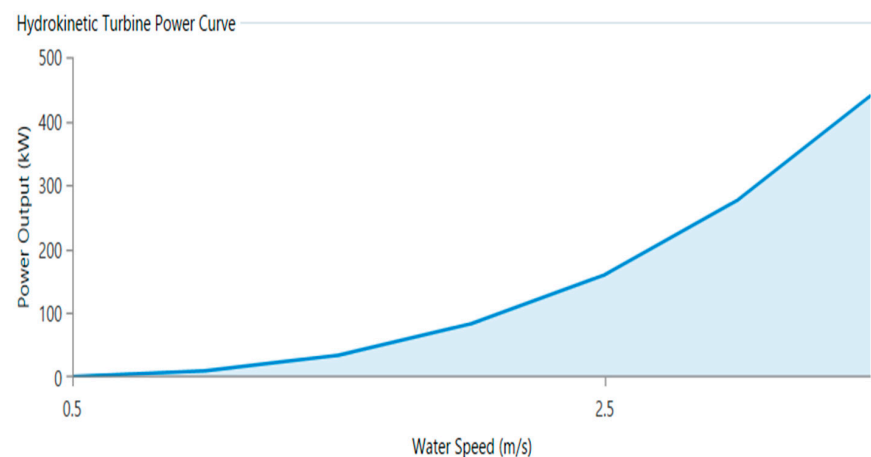


Figure 18. The water speed.

6.2. Tidal Turbine

The Schottel (Schottel, Spray, Germany) 54 kW bidirectional turbine parameters indicated in Table 6 are selected for their low cut-in speed and their capability to meet peak load demands while supplying a significant portion of the peak load at minimal purchase, installation, and maintenance costs [58]. The capital costs, including installation, wiring, and mounting, are \$54,000, with replacement costs also at \$54,000 [58], an annual maintenance cost of \$2700, and a lifespan of 10 years [58]. The power curve and generation percentage at 0.5 m/s intervals, shown in Figure 18, are described by Equation (12):

$$P_{tidal} = 2.0978V^3 + 2.7163V^2 - 1.7909V - 0.2358 \text{ for } V \leq 2.5 \quad (12)$$

Table 6. The power curve of the Schottel 54 kW tidal turbine.

Tidal Turbine	Schottel (54 kW)
Model name	Schottel
Rated power	54 kW
Cut-in tidal speed	0.7 m/s
Cut-out tidal speed	4.6 m/s
Rated power at	2.6 m/s
Swept area	7.06 m ²
Rotor diameter	3 m
Capital cost	\$54,000
Replacement cost	\$54,000
O&M	\$2700
Lifetime	20 years

6.3. Battery Energy Storage

Energy storage plays a critical role in integrated systems that generate power from diverse renewable sources. To ensure dependable performance and cost-effective operation, a standard 12-volt lead-acid battery with 1 kWh of storage capacity is recommended [59]. Detailed specifications in Table 7 outline capital and replacement costs of \$200 each, an annual maintenance cost of \$20, and a lifespan of 10 years [58].

Table 7. Battery Components.

Battery	Values
Model name	Generic 1 kWh Lead Acid
Nominal capacity	1
Roundtrip efficiency	80%
Min State of charge	40%
Float life	5 years
Max Charge rate	1
Max Charge current	16.7 A
Capital cost	\$200
Replacement cost	\$200
O&M cost	\$20
Lifetime throughput	9600 kWh

6.4. Converter

A standard system converter is proposed as a cost-effective alternative to bidirectional converters to convert the AC output of the generator to DC. The HOMER (Homer Energy, Boulder, CO, USA) optimizer optimizes the converter's size. Capital and replacement costs are both \$54,000, with an annual maintenance cost of \$1500 [40]. The converter operates at 90% efficiency and has a lifespan of 15 years.

6.5. Controller

A load-following (LF) controller is selected for its ability to precisely match power generation to demand requirements. Capital and replacement costs for the controller are \$200 each, with an annual maintenance cost of \$5.00. The controller is designed to operate efficiently over a lifespan of 25 years [40].

7. Results and Discussion

Seasonal variations play a critical role in influencing electricity demand at the West Bank site. The analysis indicates that demand peaks during the winter months of June and July, primarily due to increased heating requirements and reduced daylight hours, which result in higher electricity consumption. Conversely, during warmer months, the demand is generally lower as heating needs diminish. Accurately assessing these seasonal fluctuations is essential for the design of turbine systems capable of efficiently meeting the variable demand throughout the year.

This section examines off-grid hybrid systems—comprising tidal, wind, storage, and diesel generator turbines—to meet the electricity demand at the West Bank, East London site. Figures 9 and 10 present an analysis of the average hourly and monthly loads, while Figure 11 visually depicts this data. The peak hourly load is recorded at 597.76 kW. The HOMER software analysis, as shown in Figure 15, reveals that the average daily load fluctuates around 4500 kWh/day. The corresponding graph illustrates the load variations throughout the year, with June and July identified as periods of peak demand. After applying a load factor of 0.69, the total annual demand is computed to be 3,187,017 kWh. The power output variability of the tidal turbine and wind turbine is indicated in Figures 19 and 20.

Figure 21 shows the power demand fluctuations over three days, comparing the outputs from a tidal turbine, a wind turbine, and a diesel generator. The red line represents the total power generated by the system, while the green line indicates the power demand. The data suggest that to decrease the West Bank coastal area's reliance on diesel generators, it is essential to implement a hybrid system with multiple wind and tidal turbines.

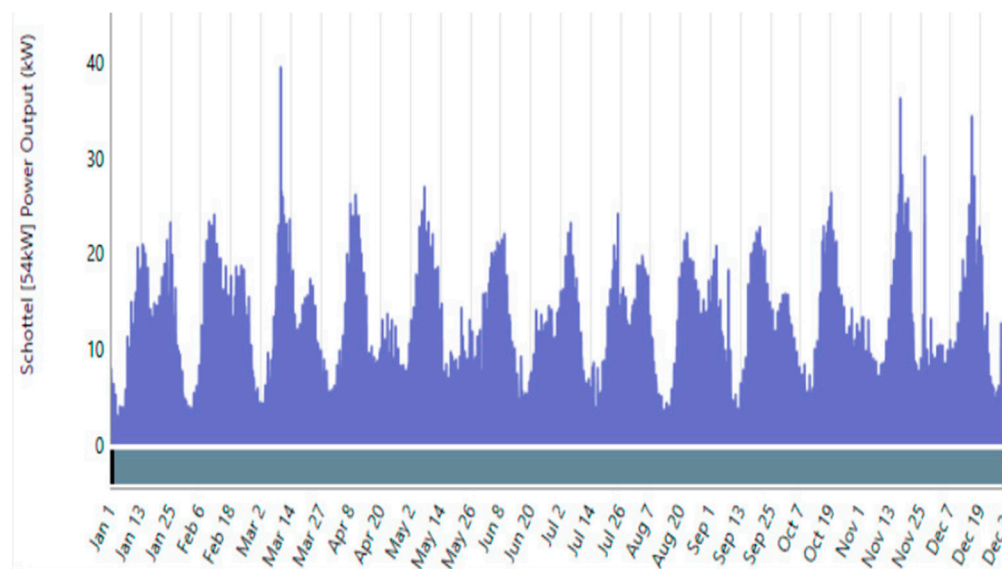


Figure 19. Power output variability for the tidal turbine at the West Bank East London site.

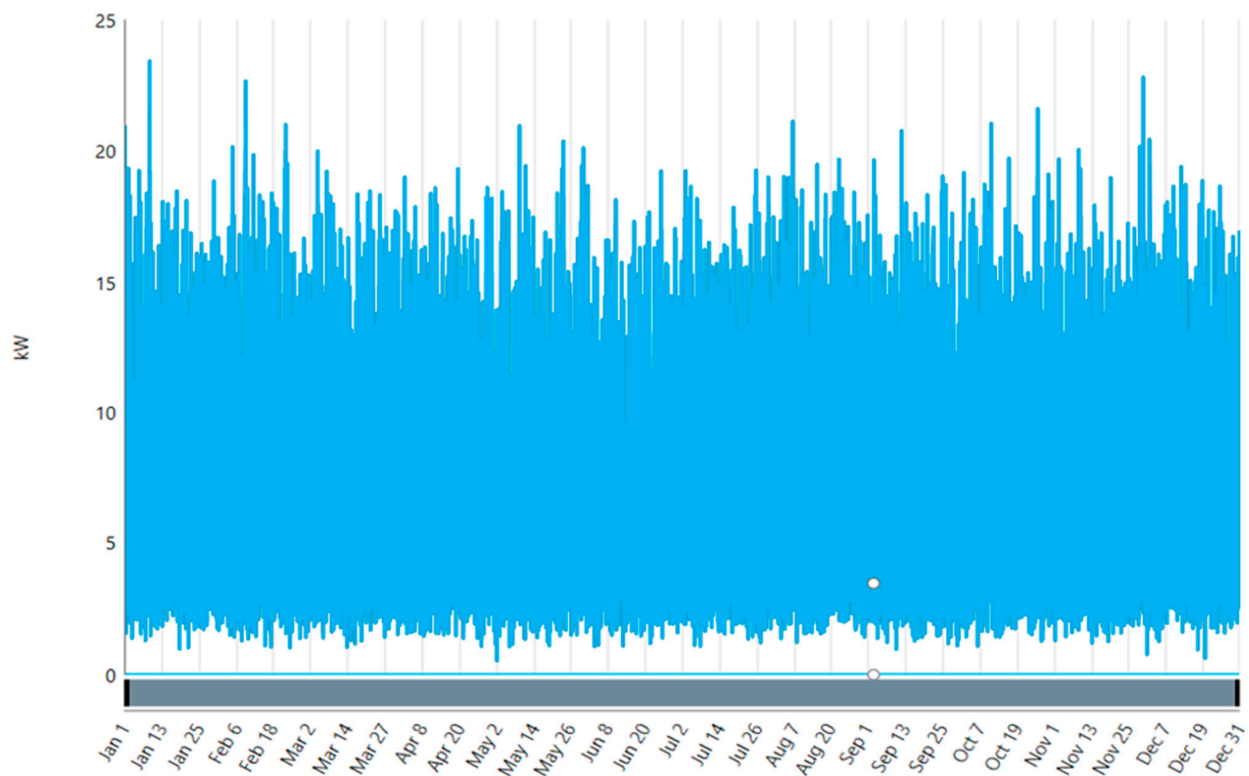


Figure 20. Power output variability for a West Bank East London site wind turbine.

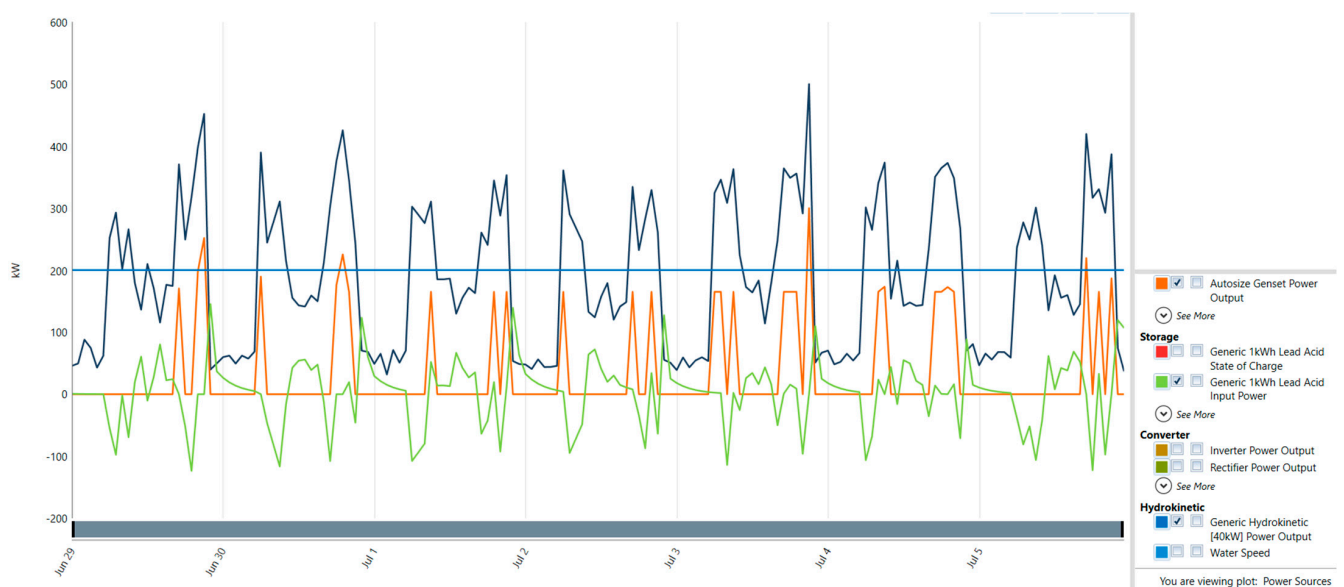


Figure 21. Fluctuations in power production and consumption over three days at West Bank East London.

7.1. Financial Analysis Results

Figure 16 illustrates how HOMER software evaluates and optimizes various configurations of turbines and generators to manage demand fluctuations throughout the year. The scenarios analyzed include different combinations of wind turbines (W), tidal turbines (T), energy storage systems (S), and diesel generators (G). Sections 7.1–7.3 delve into the outcomes of these configurations, with a focus on power generation efficiency, cost-effectiveness, and environmental emissions.

Operating costs, which are crucial to the analysis, encompass all expenses related to preliminary studies, project development, environmental impact assessments, permitting procedures, and project management. These costs can vary significantly depending on the project's location, the national policies in place, and the scale of the installation. Consequently, a broad range of cost estimates is presented, as shown in the subsequent figures.

This study employs HOMER software to conduct a thorough financial analysis of a hybrid wind-tidal energy storage system. The analysis is based on predefined parameters and assumptions in Table 8, providing a comprehensive understanding of the system's economic viability.

Table 9 presents the optimization results for an alternative scenario that balances wind and tidal generation with a configuration of 2 wind turbines and 5 tidal turbines in rank (b) (2W + 5T), achieving a renewable fraction of 52.7%. However, this configuration results in higher capital costs, operating costs, and Net Present Costs (NPC) compared to the configurations of 1 wind turbine and 5 tidal turbines in rank (d), (h), (j) (1W + 5T). As a result, the Cost of Energy (COE) is also higher. Table 10 summarizes the total electricity production and the percentage contributions from each energy source, demonstrating that in the storage + 20T scenario, two-thirds of the production is sourced from renewables. The 5 tidal turbines (2 + 5T) are more cost-effective in terms of installation and operation. Table 9 in rank (b), (d), (h), (j), and (k) present the optimization results for each scenario, showing renewable fractions of 30.2%, 32.5%, 57.1%, and 61.1%, respectively. A renewable fraction of 61.1% indicates that 61.1% of the electricity is generated from wind and tidal turbines, while the remaining electricity is provided by the diesel generator. Hybrid systems with a renewable fraction exceeding 66% are generally more profitable. The 1W + 5T and 2W + 5T scenarios nearly meet this criterion and have lower operating costs, fuel costs, and Net Present Costs (NPC) compared to the 20W + 5T scenarios.

Table 8. Economic Analysis of Balanced Alternative of the Hybrid Energy Systems: Cost Summary and NPC Component Breakdown.

Component	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Savage (\$)	Total (\$)
Auto size Genset	\$300,000	\$790,403.24	\$822,102.75	\$2,273,714.20	\$45,108.53	\$4,171,111.66
Generic 1 kwh Lead Acid	\$268,200	\$436,780.18	\$11,283.20	\$0.00	\$0.00	\$819,263.39
Gen Hydro (40 kw)	\$70,000	\$60,913.97	\$0.00	\$0.00	\$8154.95	\$122,759.01
Homer Load Following	\$200	\$0.00	\$255.67	\$0.00	\$0.00	\$455.67
System Converter	\$44,578.37	\$18,601.00	\$0.00	\$0.00	\$3154.95	\$59,717.13
XANT M-21 (100 KW)	\$300,000	\$77,951.18	\$127,833.56	\$0.00	\$43,687.24	\$462,097.50
System	\$1,012,978.37	\$1,384,649.57	\$1,064,475.18	\$2,273,714.20	\$100,412.96	\$5,635,404.36

Table 9. The optimization results for an alternative.

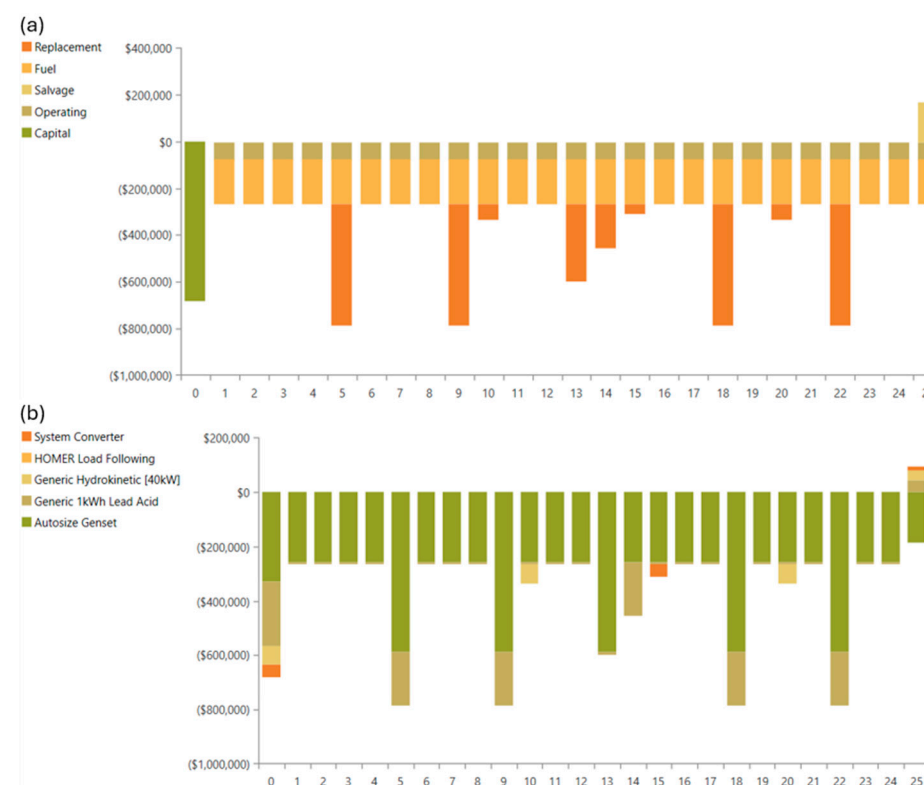
Rank	Architecture					COST			
	M-21 (kW)	Gen (kW)	1 kwh LA	HK40	Converter kW	Dispatch	NPC (\$)	COE (\$)	Operat-Cost
a		660	787	5	149	LF	\$5.43 M	\$0.259	\$371,620
b	2	660	911	5	321	LF	\$4.74 M	\$0.226	\$263,918
c		660	2132	5	428	LF	\$3.48 M	\$0.171	\$136,140
d	1	660	1088	5	163	CC	\$8.23 M	\$0.392	\$582,768
e		660	7192	5	427	CC	\$5.49 M	\$0.271	\$190,567
f		660	787	5	149	LF	\$5.43 M	\$0.259	\$371,620
h	1	660	894	5	149	LF	\$5.64 M	\$0.268	\$361,617
i		660		5		CC	\$6.97 M	\$0.332	\$513,742
j	1	660		5	18.7	CC	\$7.37 M	\$0.351	\$521,439
k	20		7192	5	455	CC	\$13.9 M	\$0.690	\$435,050

Table 10. Economic metrics of the optimized system.

Descriptions	Values
Simple payback:	2.05 yr
Return on Investment:	46.1%
Internal Rate of Return:	57.2%
Net Present Value:	\$1.54 M
Capital Investment:	\$281,063
Annualized Savings:	\$142,122

It proposes adding 788 kWh of battery capacity. This would reduce your operating costs to \$371,620/yr. The investment has a payback of 2.05 years and an IRR of 57.2%.

Figure 22a,b illustrate that increasing the number of tidal turbines raises the system's capital, operating, replacement, and salvage costs. Despite this, the overall fuel cost remains unchanged. Additionally, the construction expenses are higher due to the need for extra floating platforms and cables required for the additional tidal turbines.

**Figure 22.** The economic and component chronological cash flow.

Based on the results in Table 9 and Figure 23, a modest investment can yield long-term societal benefits, with a payback period of less than 7 years over a 25-year project lifespan. Technological advancements and improved materials may extend the lifespan of wind and tidal turbines beyond 25 years. Consequently, using tidal stream turbines with a 30–40-year lifespan could significantly impact future economic calculations. Additionally, emerging tidal technologies can further enhance economic outcomes by reducing the payback period.

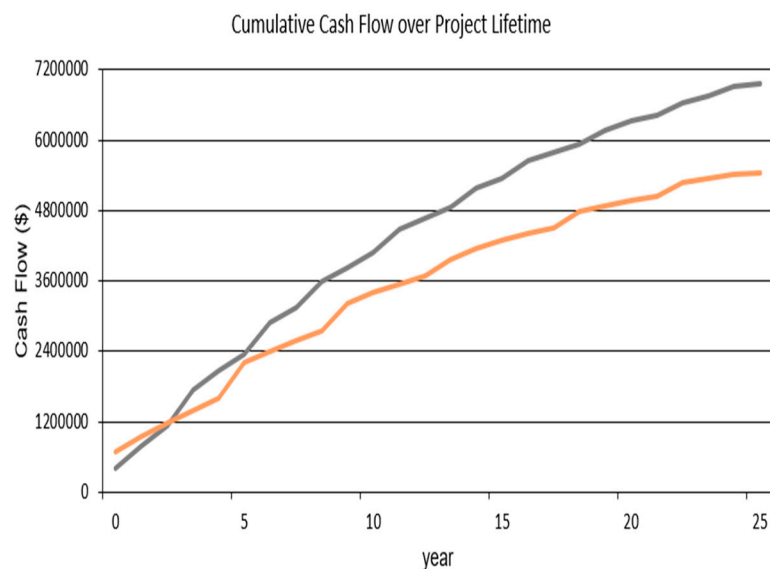


Figure 23. A simple payback period is based on cumulative cash flow throughout the project's lifetime.

7.2. Analysis of Power Generation Results

Figure 24 presents the average output, operational hours, and total production for each scenario. Increasing the number of wind and tidal turbines leads to a reduction in both the operational hours and the total output of diesel generators. In all scenarios, the total power production surpasses the West Bank coastal area's annual demand of 1,260,332 kWh.



Figure 24. Monthly and yearly average output power for (a) 5T + Gen, (b) 1W + 20T, (c) Gen + 20T.

This microgrid requires 7200 kWh/day and has a peak of 300 kW. In the proposed system, the following generation sources serve the electrical load.

Figure 23 presents the average power output for different scenarios. West Bank East London coastal site has an average power demand of 4500 kW for 150 community houses, which is satisfied by all scenarios evaluated. Particularly noteworthy are scenarios involving 1 wind turbine and 5 tidal turbines, as well as 1 wind turbine and 20 tidal turbines, both of which surpass the peak demand of 597.96 kW with their average outputs. Further analyses of the optimal result from Section 1 are detailed as follows

Over time, this initial cost is offset by the system's positive salvage value. The generator consumed a total of 224,316 L of fuel over the study period, representing the aggregate amount of diesel used for electricity production indicated in Figure 25. On average, the generator uses 615 L of fuel per day, calculated by dividing the total fuel consumption by the number of days in the study period. This provides insight into the daily fuel requirement for the generator's operation. Additionally, the generator consumes an average of 25.6 L of fuel per hour, derived by dividing the daily average fuel consumption by 24 h. This value indicates the generator's fuel efficiency and typical hourly fuel usage.

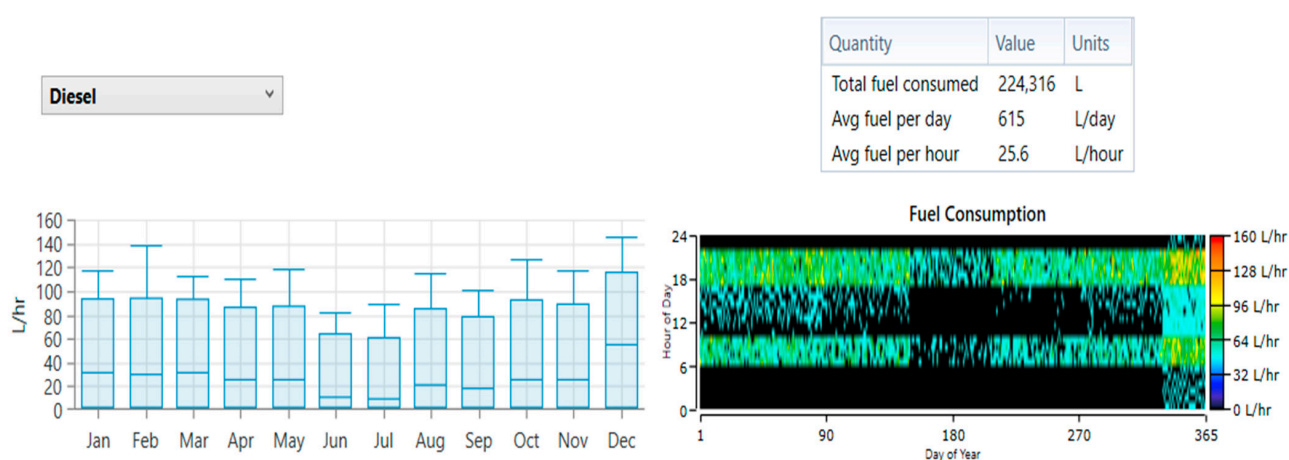


Figure 25. Diesel generator fuel consumption.

In summary, these values offer a detailed overview of the generator's fuel usage, helping to understand its operational efficiency and the overall fuel costs involved in running the generator over the given period.

Figure 26 and Tables 11 and 12 provide a detailed overview of the generator's power output values. Electrical Production: The generator produces a total of 783,387 kWh of electrical energy per year, representing the annual electricity generated by the diesel generator. The mean Electrical Output is the generator's average electrical output of 226 kW. This figure is derived by dividing the total annual electrical production by the number of hours in a year (783,387 kWh/year, 8760 h/year).

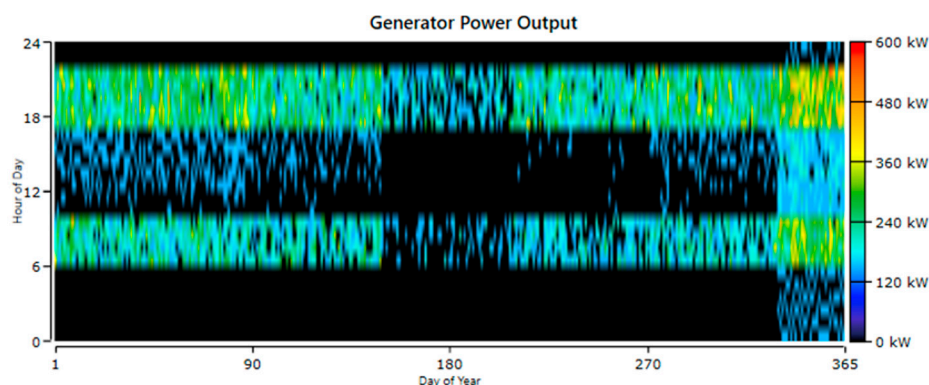


Figure 26. Generator power output.

Table 11. Generator Power Output Parameters.

Quantity	Value	Units
Hours of Operation	3469	Hrs/yr
Number of Starts	1124	Starts/yr
Operation Life	4.32	yr
Capacity Factor	13.5	%
Fixed Generation Cost	51.5	\$/h
Marginal Generation Cost	0.201	\$/kWh
Electrical Production	783,387	kWh/yr
Mean Electrical Output	226	kW
Minimum Electrical Output	165	kW
Maximum Electrical Output	565	kW
Thermal Production	569,551	kWh/yr
Mean thermal output	164	kW

Table 12. Generator Power Output Parameters.

Quantity	Value	Units
Fuel consumption	224,316	L
Specific fuel consumption	0.286	L/kWh
Fuel Energy Input	2,207,266	kWh
Mean Electrical Efficiency	35.5	%

Minimum Electrical Output is the generator’s minimum electrical output of 165 kW, indicating the lowest level of electrical power it produces at any point during its operation. Maximum electrical output: The generator’s maximum electrical output is 565 kW, representing the highest level of electrical power it can produce.

Thermal production: The generator also produces 569,551 kWh of thermal energy per year, representing the total amount of heat generated annually, which can be utilized for heating purposes. The mean Thermal Output is the average thermal output of the generator, which is 164 kW. This value is calculated by dividing the total annual thermal production by the number of hours a year (569,551 kWh/year, 8760 h). Figure 27 illustrates the battery bank’s state of charge, starting at 30%.

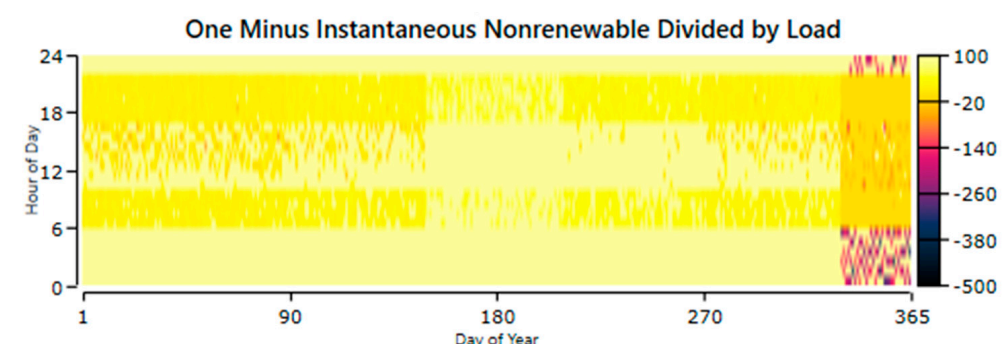
**Figure 27.** One minus instantaneous non-renewable divided by the load.

Figure 27 provides an analysis of the instantaneous renewable output of the load, offering key insights into the renewable energy contributions and their impact on overall energy production. Total Renewable Production Divided by Load—this metric indicates that 61.1% of the total load demand is met by renewable energy sources. It highlights the substantial contribution of renewable energy in covering the community’s electricity needs, demonstrating the effectiveness of integrating renewable sources into the power system. Total Renewable Generator: This value represents that 56.1% of the total energy generated by the system comes from renewable sources. It underscores the dominance of

renewable energy in the generation mix and reflects the system’s reliance on sustainable energy production.

One Minus Total Non-renewable Production Divided by Load: This metric, calculated as $1 - (\text{total non-renewable production} / \text{load})$, shows that 52.3% of the load is met without relying on non-renewable energy sources. It illustrates the extent to which the system reduces dependence on non-renewable energy, highlighting its environmental benefits and contribution to sustainability.

In the context of a research paper, these parameters from the simulation or analysis tool for the battery system indicated in Tables 13 and 14, and Figure 28 provide crucial insights into its operational characteristics and economic feasibility. Here’s how each parameter contributes to understanding the battery system’s performance. The Battery Bank State of Charge Functional Characteristics: This parameter elucidates how the battery’s state of charge (SOC) evolves, reflecting its ability to store and discharge energy efficiently under varying operational conditions. Understanding SOC dynamics is critical for optimizing battery usage and ensuring a reliable energy supply. Anatomy 2.52 h: While not fully defined in the context provided, it likely pertains to a specific operational or design aspect of the battery system, potentially related to its discharge or charge profile over a defined period.

Storage Wear Cost is quantified at \$0.3335 per kWh; this parameter assesses the economic impact of wear and tear on the battery due to its usage. It underscores the importance of lifecycle cost considerations in evaluating the economic viability of battery deployment. Nominal Capacity: At 473 kWh, this parameter denotes the total energy storage capability of the battery system, influencing its ability to meet energy demands and integrate renewable energy sources effectively.

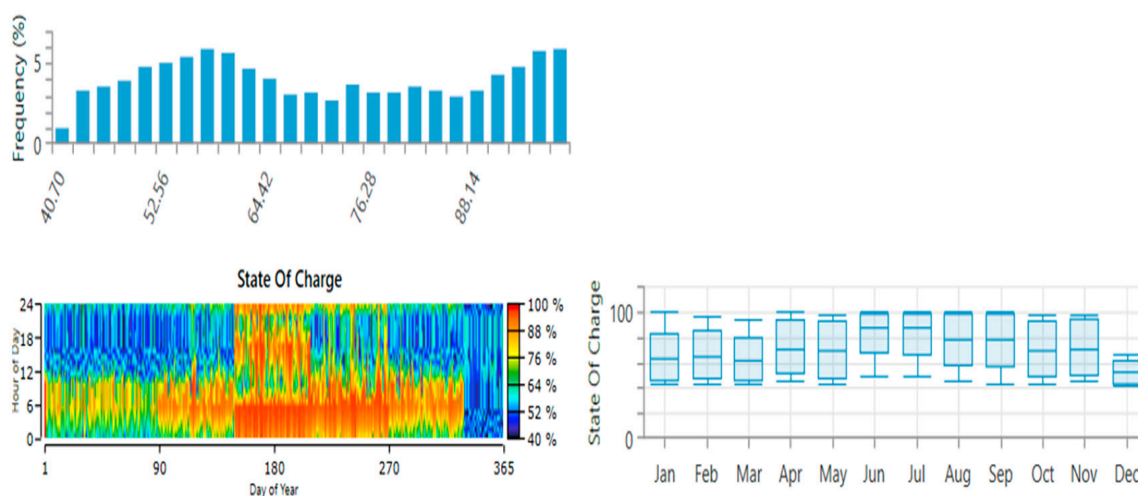


Figure 28. Battery Bank State of Charge.

Table 13. Battery Bank State of Charge parameters.

Quantity	Value	Units
Batteries	787	Qty.
String Size	1.00	batteries
Strings in Parallel	787	strings
Bus Voltage	12.0	v
Autonomy	2.52	h
Storage Wear Cost	0.335	\$/kWh
Nominal Capacity	788	kWh
Usable Nominal Capacity	473	kWh
Lifetime Throughput	629,600	kWh
Expected Life	4.36	yr

Table 14. Battery Bank State of Charge parameters.

Quantity	Inverter	Units
Average Energy Cost	161,060	\$/kWh
Energy In	129,205	kWh/yr
Energy Output	0	kWh/yr
Storage Depletion	129	kWh/yr
Losses	9.39	kWh/yr
Annual Throughput	144,456	kWh/yr

Lifetime Throughput is 629,600 kWh processed over its operational lifespan; this metric quantifies the total energy handled by the battery, offering insights into its long-term performance and durability. Expected Life is projected at 4.36 years, and this parameter estimates the operational duration of the battery system, guiding decisions on maintenance schedules, replacement planning, and overall economic return on investment.

The average energy cost is noting \$0 per kWh. This parameter implies that the cost of energy storage and discharge does not contribute to an additional financial burden beyond initial setup costs, affecting the overall economic feasibility positively. Energy in and Out Annual figures of 161,060 kWh and 129,205 kWh, respectively, illustrate the energy flow into and out of the battery system annually. These metrics gauge operational efficiency and energy supply reliability under real-world conditions.

Storage Depletion and Losses are these parameters—399 kWh/year and 32,254 kWh/year—highlight the amount of energy drawn from the battery annually and the losses incurred within the system, influencing overall efficiency and operational costs.

Annual Throughput: Summing to 1,444,456 kWh/year, this metric reflects the total energy processed annually by the battery system, encompassing both input and output energy.

In the context of our study on the standalone offshore hybrid wind-tidal storage energy system, the inverter and rectifier parameters indicated in Table 15 and Figure 29 provide essential insights into their operational efficiency and capacity. The inverter, with a capacity of 149 kW, handles an average output of 14 kW and peaks at 129 kW, indicating its ability to manage variable power demands efficiently. Its capacity factor of 9.39% and 2039 h of operation per year reflect moderate utilization levels. The annual energy output of 122,745 kWh and input of 129,205 kWh, with losses amounting to 6460 kWh, underscores the importance of minimizing conversion losses to enhance overall system efficiency.

Similarly, the rectifier, also with a capacity of 149 kW, demonstrates a higher average output at 18 kW and a capacity factor of 12.3%, operating for 4426 h annually. Its energy output of 161,060 kWh and input of 169,537 kWh, alongside losses of 8477 kWh, highlight its critical role in converting and storing energy effectively. These parameters collectively inform the design and operational strategies to optimize the performance and reliability of the energy system.

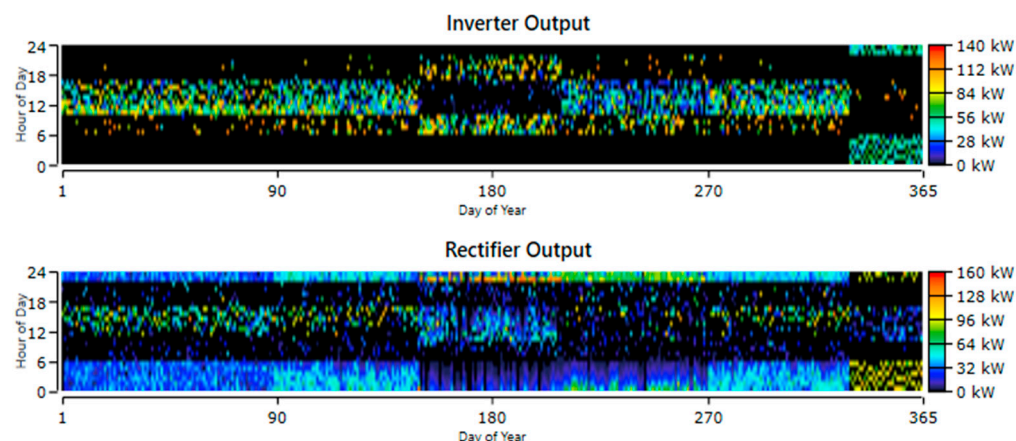
**Figure 29.** Inverter and rectifier out.

Table 15. Inverter and rectifier out parameters.

Quantity	Inverter	Rectifier	Units
Capacity	149	149	kW
Mean Output	14.0	18.4	kW
Minimum Output	0	0	kW
Maximum Output	129	149	kW
Capacity Factor	9.39	12.3	%

In summary, understanding the detailed functioning of the inverter and rectifier is vital for ensuring efficient energy management and maximizing the benefits of integrating renewable energy sources within the hybrid system.

Table 15 and Figure 29 highlight the outputs of the inverter and rectifier, with the inverter output showing higher oscillations than the rectifier.

Table 16 illustrates the surplus electric energy generated by a standalone microgrid equipped with wind and tidal storage, highlighting its capacity to meet a significant portion of the 24-h power demand. In contrast, the diesel generator can only cover a small fraction of the load. The table provides details on excess energy, unmet loads, and capacity shortages within the system. Excess energy refers to the power generated but not utilized by any load. The analysis of the microgrid's components offers valuable insights into the system's design, economic viability, and operational efficiency. The microgrid's rated capacity of 1080 kW indicates its nominal power output, while the capital cost of approximately \$58,000 represents the initial investment required. Engineered for continuous and reliable power generation, the system is designed to operate 8760 h per year.

Table 16. A surplus of generated electricity.

Quantity	Value	Units
Excess Electricity	17,913,435	Kg/yr
Unmet Electric	0	Kg/yr
Capacity Shortage	0	Kg/yr
Renewable Fraction	100	%
Max. Renew. Penetration	321,133	%

The levelized cost of \$0.00081/kWh underscores the cost-effectiveness of this energy production method, ensuring affordable electricity over the system's lifespan. The exceptionally high capacity factor of 141% warrants further investigation but suggests that the system might be oversized or there are unique operational conditions leading to such high utilization.

With a total annual production of 13,334,303 kWh, the microgrid component demonstrates significant energy generation capability, supporting substantial energy demands. The annual maintenance cost of approximately \$2900 is a modest investment to keep the system in optimal condition.

The maximum output of 3200 kW highlights the system's robustness in handling peak power demands. However, the hydrokinetic penetration level of 5883 is unusually high, indicating an overwhelming reliance on hydrokinetic energy. This could potentially exceed local demand, suggesting energy export opportunities or possible reporting inaccuracies.

In conclusion, these parameters provide a comprehensive overview of the microgrid component's performance, highlighting its capacity, economic efficiency, operational longevity, and energy production dominance, particularly from hydrokinetic sources. These insights are crucial for optimizing the design and operation of hybrid energy systems to ensure sustainable and cost-effective power supply.

The microgrid wind-tidal storage system produces 17,913,435 kW/yr of excess electricity with zero unmet load and capacity shortages. It achieves a 100% renewable fraction and a maximum renewable penetration of 321.133%. This system ensures a reliable, sustainable

energy supply, significantly reduces emissions, lowers operational costs, and provides economic benefits by potentially selling excess electricity back to the grid.

7.3. Environmental Emissions

The fuel consumption of diesel generators primarily drives annual environmental emissions. Table 16 compares yearly emissions of various pollutants across different study locations in Bangladesh. With renewable energy sources and battery storage, the diesel generator solely meets the load demand, leading to higher fuel consumption and increased environmental emissions. Figure 18 illustrates the annual diesel fuel consumption and corresponding CO₂ emissions for different locations.

The emission proportions from the wind-tidal microgrid system are significantly lower than those from fossil fuel-based systems, with annual emissions of 587,172 kg of CO₂, 3701 kg of CO, 162 kg of unburned hydrocarbons, 22.4 kg of particulate matter, 1438 kg of SO₂, and 3477 kg of NO_x. These reductions highlight the environmental benefits and sustainability of integrating wind and tidal energy. The wind-tidal combination is the optimal choice for minimizing environmental impact and ensuring cleaner air in the microgrid, demonstrating a substantial advantage over traditional fossil fuel systems. The emissions contribute to climate change and pose significant health risks.

The environmental emissions of the Tidal/Wind/Diesel/Battery system (Table 17) in the studied areas are comparable to those of a microgrid that combines wind, diesel, and battery storage. However, due to the intermittency of wind energy, the system often relies on the diesel generator, as wind energy contributes minimally to the total load demand. In contrast, the hybrid Tidal/Diesel/Battery system achieves a significant reduction in environmental emissions compared to the diesel-only system.

Table 17. Emissions of pollutants and fuel consumption.

Quantity	Value	Units
Carbon Dioxide	587,172	Kg/yr
Carbon Monoxide	3701	Kg/yr
Unburned Hydrocarbons	162	Kg/yr
Particulate Matter	22.4	Kg/yr
Sulphur Dioxide	1438	Kg/yr
Nitrogen Oxides	3477	Kg/yr

The Homer software facilitates sensitivity analysis of system parameters, particularly the wind speed and current velocity, which is essential for informed decision-making in practical applications. Figure 30 illustrates the outcomes of this sensitivity analysis, depicting power flow scenarios for independent power generation systems, combined wind power flow systems, and independent wind power generation systems from top left to bottom right. By utilizing actual wind speed and current velocity data from specific marine environments, this analysis aids in identifying the most cost-effective power generation scheme that minimizes the Net Present Cost (NPC).

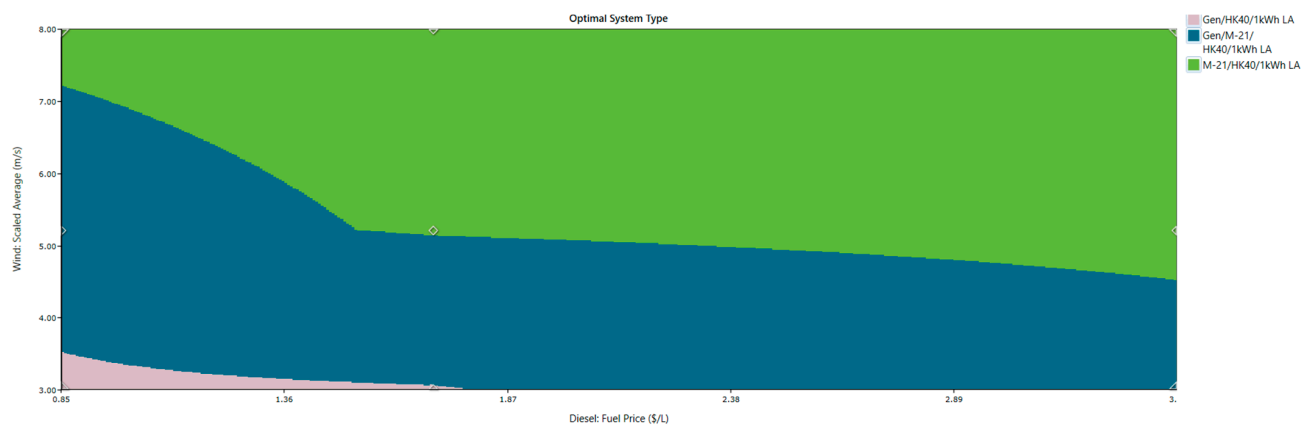


Figure 30. The outcomes of this sensitivity analysis.

8. Conclusions

Renewable energy is critical in mitigating climate change and reducing reliance on fossil fuels. Among various renewable sources, marine energy, particularly offshore wind, and tidal power, holds significant potential for sustainable power generation. This study demonstrates the feasibility and benefits of a hybrid offshore wind and tidal energy system for off-grid coastal communities in South Africa, contributing to the country's renewable energy transition and broader sustainability goals.

The integration of offshore wind, tidal energy, energy storage, and diesel backup addresses key challenges related to the intermittency and unpredictability of renewable sources. By ensuring a stable and reliable energy supply, the system significantly reduces greenhouse gas emissions while minimizing environmental impacts. Optimization via HOMER software has shown that this hybrid configuration not only improves energy reliability but also increases the renewable energy fraction and reduces the net present cost—key factors in advancing sustainability.

The study underscores that hybrid systems can maximize resource utilization and enhance energy security in coastal regions. By leveraging the complementary characteristics of wind and tidal energy, the system mitigates the limitations of single-resource dependency, creating a more resilient and adaptable energy infrastructure. Moreover, incorporating externalities into the cost calculations reveals that renewable energy, when accounting for environmental costs, becomes more competitive with conventional sources, further closing the cost gap.

Future research and development in hybrid converter technologies and large-scale deployment will be essential to fully harness the potential of marine energy systems. This research provides a framework for the successful implementation of hybrid wind and tidal systems, aligning with South Africa's sustainability initiatives while offering a scalable model for other coastal regions. In doing so, it supports the global shift towards cleaner, more resilient, and sustainable energy systems.

Author Contributions: L.M.K., A.R. and E.O. contributed to the study's planning, idea generation, and information field analysis, as well as literature review and conceptualization. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not Applicable.

Informed Consent Statement: Not Applicable.

Data Availability Statement: All data supporting the results are included within the article, and no additional source data are necessary.

Acknowledgments: I would like to express our sincere gratitude to the Cape Peninsula University of Technology (CPUT) for their generous funding and support throughout this research. Our deepest thanks also go to Raji and Efe for their invaluable guidance, expertise, and encouragement, which greatly contributed to the success of this study. Their insights and feedback were crucial in shaping the direction of this research.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Yimen, N.; Tchotang, T.; Kanmogne, A.; Idriss, I.A.; Musa, B.; Aliyu, A.; Okonkwo, E.C.; Abba, S.I.; Tata, D.; Meva, L.; et al. Optimal Sizing and Techno-Economic Analysis of Hybrid Renewable Energy Systems—A Case Study of a photo-voltaic/wind/battery/diesel system in Fanisau, Northern Nigeria. *Processes* **2020**, *8*, 1381. [\[CrossRef\]](#)
2. Majdi Nasab, N.; Kilby, J.; Bakhtiaryfard, L. Case study of a hybrid wind and tidal turbines system with a microgrid for power supply to a remote off-grid community in New Zealand. *Energies* **2021**, *14*, 3636. [\[CrossRef\]](#)
3. Baykov, A.; Dar'enkova, A.; Kurkin, A.; Sosnina, E. Mathematical modelling of a tidal power station with diesel and wind units. *J. King Saud Univ. Sci.* **2019**, *31*, 1491–1498. [\[CrossRef\]](#)
4. Ren, Z.; Li, H.; Li, W.; Zhao, X.; Sun, Y.; Li, T.; Jiang, F. Reliability Evaluation of Tidal Current Farm Integrated Generation Systems Considering Wake Effects. *IEEE Access* **2018**, *6*, 52616–52624. [\[CrossRef\]](#)
5. Sousounis, M.C.; Shek, J.K.H. Assessment of pulsating torque mitigation control strategy through tidal turbine emulation. *J. Eng.* **2019**, *2019*, 5059–5063. [\[CrossRef\]](#)
6. Mohammed, O.H.; Amirat, Y.; Benbouzid, M.E.H.; Feld, G. Optimal design and energy management of a hybrid power generation system based on wind/tidal/PV sources: Case study for the ouessant French Island. In *Green Energy and Technology*; Springer: Cham, Switzerland, 2017; pp. 381–413. [\[CrossRef\]](#)
7. Abdalla, A.N.; Nazir, M.S.; Tao, H.; Cao, S.; Ji, R.; Jiang, M.; Yao, L. Integration of energy storage system and renewable energy sources based on artificial intelligence: An overview. *J. Energy Storage* **2021**, *40*, 102811. [\[CrossRef\]](#)
8. Suchitra, D.; Rajarajeswari, R.; Dasgupta, J. Optimization of hybrid renewable energy system using homer. *Int. J. Recent Technol. Eng.* **2019**, *8*, 2542–2550. [\[CrossRef\]](#)
9. Liu, Y.; Du, W.; Xiao, L.; Wang, H.; Bu, S.; Cao, J. Sizing a Hybrid Energy Storage System for Maintaining Power Balance of an Isolated System with High Penetration of Wind Generation. *IEEE Trans. Power Syst.* **2016**, *31*, 3267–3275. [\[CrossRef\]](#)
10. Jacob, A.S.; Banerjee, R.; Ghosh, P.C. Sizing of hybrid energy storage system for a PV based microgrid through design space approach. *Appl. Energy* **2018**, *212*, 640–653. [\[CrossRef\]](#)
11. Arabali, A.; Ghofrani, M.; Etezadi-Amoli, M.; Fadali, M.S. Stochastic performance assessment and sizing for a hybrid power system of Solar/Wind/Energy Storage. *IEEE Trans. Sustain. Energy* **2014**, *5*, 363–371. [\[CrossRef\]](#)
12. Said, H.A.; García-Violini, D.; Ringwood, J.V. Wave-to-grid (W2G) control of a wave energy converter. *Energy Convers. Manag.* **2022**, *14*, 100190. [\[CrossRef\]](#)
13. Lafoz, M.; Blanco, M.; Belouqui, L.; Navarro, G.; Moreno-Torres, P. Dimensioning methodology for energy storage devices and wave energy converters supplying isolated loads. *IET Renew. Power Gener.* **2016**, *10*, 1468–1476. [\[CrossRef\]](#)
14. Forestieri, J.N.; Farasat, M. Integrative Sizing/Real-Time Energy Management of a Hybrid Supercapacitor/Undersea Energy Storage System for Grid Integration of wave energy conversion systems. *IEEE J. Emerg. Sel. Top. Power Electron.* **2020**, *8*, 3798–3810. [\[CrossRef\]](#)
15. Akhtari, M.R.; Baneshi, M. Techno-economic assessment and optimization of a hybrid renewable co-supply of electricity, heat and hydrogen system to enhance performance by recovering excess electricity for a large energy consumer. *Energy Convers. Manag.* **2019**, *188*, 131–141. [\[CrossRef\]](#)
16. Sharma, S.; Sood, Y.R.; Sharma, N.K.; Bajaj, M.; Zawbaa, H.M.; Turkey, R.A.; Kamel, S. Modeling and sensitivity analysis of grid-connected hybrid green microgrid system. *Ain Shams Eng. J.* **2022**, *13*, 101679. [\[CrossRef\]](#)
17. Ghayoor, F.; Swanson, A.G.; Sibanda, H. Optimal sizing for a grid-connected hybrid renewable energy system: A case study of the residential sector in Durban, South Africa. *J. Energy S. Afr.* **2021**, *32*, 11–27. [\[CrossRef\]](#)
18. Mohapatra, A.; Mohapatra, S.; Bora, S.N. Hybrid renewable energy system for real-time power management techniques—A review. In Proceedings of the 2020 IEEE International Symposium on Sustainable Energy, Signal Processing and Cyber Security (iSSSC), Gunupur Odisha, India, 16–17 December 2020; pp. 1–6. [\[CrossRef\]](#)
19. Hamad, A.A.; Nassar, M.E.; El-Saadany, E.F.; Salama, M.M.A. Optimal Configuration of Isolated Hybrid AC/DC Microgrids. *IEEE Trans. Smart Grid* **2019**, *10*, 2789–2798. [\[CrossRef\]](#)
20. Sun, P.; Teng, Y.; Leng, O.; Chen, Z. Stability control method for hybrid AC-DC transmission systems considering cross-region multi-energy coordination. *CSEE J. Power Energy Syst.* **2020**, *7*, 753–760. [\[CrossRef\]](#)
21. Kajela, D.; Manishahia, M.S. Optimization of Renewable Energy Systems: A Review. *Int. J. Sci. Res. Sci. Technol.* **2017**, *3*, 769–795.
22. Shirai, Y.; Minamoto, S.; Yonemura, K.; Rahman, M.L. Output power control of hybrid off-shore-wind and tidal turbine generation system with battery storage system. In Proceedings of the 19th International Conference on Electrical Machines and Systems (ICEMS), Chiba, Japan, 13–16 November 2016; pp. 1–6.

23. Raza, M. Offshore Grid Control of Voltage Source Converters For Integrating Offshore Wind Power Plants. Ph.D. Thesis, Universitat Politècnica de Catalunya, Barcelona, Spain, 2017. Available online: <http://hdl.handle.net/2117/114452> (accessed on 20 June 2024).
24. Mrcela, I.; Sumina, D.; Sacic, F.; Barisa, T. A wind turbine two level back-to-back converter power loss study. In Proceedings of the 2016 IEEE International Power Electronics and Motion Control Conference (PEMC), Varna, Bulgaria, 25–28 September 2016; pp. 308–314. [\[CrossRef\]](#)
25. Kangaji, L.; Orumwense, E.; Abo-al-ez, K.M. Modelling and performance analysis of grid connected tidal energy generation system. In Proceedings of the AIUE Proceedings of the 2nd Energy and Human Habitat Conference 2021, Cape Town, South Africa, 8 September 2021; pp. 1–6.
26. Zhou, K.; Cherukuru, N.; Sun, X.; Calhoun, R. Wind gust detection and impact prediction for wind turbines. *Remote Sens.* **2018**, *10*, 514. [\[CrossRef\]](#)
27. Ni, K.; Hu, Y.; Liu, Y.; Gan, C. Performance analysis of a four-switch three-phase grid-side converter with modulation simplification in a doubly-fed induction generator-based wind turbine (DFIG-WT) with different external disturbances. *Energies* **2017**, *10*, 706. [\[CrossRef\]](#)
28. Bensalah, A.; Barakat, G.; Amara, Y. Electrical Generators for Large Wind Turbine: Trends and Challenges. *Energies* **2022**, *15*, 6700. [\[CrossRef\]](#)
29. Shawqran, A.M.; El-marhomy, A.; Attia, M.A.; Abdelaziz, A.Y.; Haes, H. Heliyon Enhancement of wind energy conversion system performance using adaptive fractional order PI blade angle controller. *Heliyon* **2021**, *7*, e08239. [\[CrossRef\]](#) [\[PubMed\]](#)
30. Rehman, S. *Wind Power Resource Assessment, Design of Grid-Connected Wind Farm and Hybrid Power System*; Elsevier: Amsterdam, The Netherlands, 2012.
31. Pérez, C.; Rivero, M.; Escalante, M.; Ramirez, V.; Guilbert, D. Influence of Atmospheric Stability on Wind Turbine Energy Production: A Case Study of the Coastal Region of Yucatan. *Energies* **2023**, *16*, 4134. [\[CrossRef\]](#)
32. McKenna, R.; Pfenninger, S.; Heinrichs, H.; Schmidt, J.; Staffell, I.; Bauer, C.; Gruber, K.; Hahmann, A.N.; Jansen, M.; Klingler, M.; et al. High-resolution large-scale onshore wind energy assessments: A review of potential definitions, methodologies and future research needs. *Renew. Energy* **2022**, *182*, 659–684. [\[CrossRef\]](#)
33. Todeschini, G. Review of tidal lagoon technology and opportunities for integration within the UK energy system. *Inventions* **2017**, *2*, 14. [\[CrossRef\]](#)
34. Blavette, A.; O'Sullivan, D.L.; Lewis, A.W.; Egan, M.G. Grid integration of wave and tidal energy. In Proceedings of the International Conference on Offshore Mechanics and Arctic Engineering—OMAE, Rotterdam, The Netherlands, 19–24 June 2011; Volume 5, pp. 749–758.
35. Jahromi, M.J.; Maswood, A.I.; Tseng, K.J. Design and evaluation of a tidal in-stream generator power port. *IEEE Syst. J.* **2013**, *7*, 723–731. [\[CrossRef\]](#)
36. Zhou, H.; Ge, S.; Qin, L. Coordinate Fault Ride-Through Strategy for Connection of Offshore Wind Farms Using Voltage Source-Converter-Based High-Voltage Direct-Current Transmission under Single Polar Fault. *Sensors* **2023**, *23*, 5760. [\[CrossRef\]](#)
37. Mousavi, G.S.M. An autonomous hybrid energy system of wind/tidal/microturbine/battery storage. *Int. J. Electr. Power Energy Syst.* **2012**, *43*, 1144–1154. [\[CrossRef\]](#)
38. Ugalde-loo, C.E.; Ekanayake, J.B.; Member, S.; Jenkins, N. State-Space Modeling of Wind Turbine Generators for Power System Studies. *IEEE Trans. Ind. Appl.* **2013**, *49*, 223–232. [\[CrossRef\]](#)
39. Mitchell, D.; Blanche, J.; Harper, S.; Lim, T.; Gupta, R.; Zaki, O.; Tang, W.; Robu, V.; Watson, S.; Flynn, D. A review: Challenges and opportunities for artificial intelligence and robotics in the offshore wind sector. *Energy AI* **2022**, *8*, 100146. [\[CrossRef\]](#)
40. Margaret Amutha, W.; Rajini, V. Techno-economic evaluation of various hybrid power systems for rural telecom. *Renew. Sustain. Energy Rev.* **2015**, *43*, 553–561. [\[CrossRef\]](#)
41. Africa, S. Trend Analysis of Climate Variability over the West Bank-East London Area, Trend Analysis of Climate Variability over the West Bank—East London Area, South Africa (1975–2011). *J. Geogr. Geol.* **2013**, *5*, 131–147. [\[CrossRef\]](#)
42. El Tawil, T.; Charpentier, J.F.; Benbouzid, M. Sizing and rough optimization of a hybrid renewable-based farm in a stand-alone marine context. *Renew. Energy* **2018**, *115*, 1134–1143. [\[CrossRef\]](#)
43. Alex, A. Tidal Stream Energy Integration with Green Hydrogen Production: Energy Management and System Optimisation. Ph.D. Thesis, Normandie Université, Caen, France, 2021.
44. Colby, J.A.; Corren, D.R.; Adonizio, M.A.; Hernandez, A.A. Advancement of a tidal energy converter mount through integrated design process and risk management. In Proceedings of the Annual Offshore Technology Conference, Houston, TX, USA, 6–9 May 2019; Volume 2019.
45. Kebede, A.A.; Coosemans, T.; Messagie, M.; Jemal, T.; Behabtu, H.A.; Van Mierlo, J.; Berecibar, M. Techno-economic analysis of lithium-ion and lead-acid batteries in stationary energy storage application. *J. Energy Storage* **2021**, *40*, 102748. [\[CrossRef\]](#)
46. Ghenai, C.; Janajreh, I. Design of Solar-Biomass Hybrid Microgrid System in Sharjah. *Energy Procedia* **2016**, *103*, 357–362. [\[CrossRef\]](#)
47. Grobler, J.-H.; Carter-Brown, C.; Calitz, J.; Roro, K. *Draft Final Report Potential of Offshore Wind Energy as a Power Source for South Africa*; CSIR Smart Places Cluster Energy Centre: Pretoria, South Africa, 2022.
48. Mohammed, O.H.; Amirat, Y.; Benbouzid, M. Particle swarm optimization of a hybrid wind/tidal/PV/battery energy system. Application to a remote area in Bretagne, France. *Energy Procedia* **2019**, *162*, 87–96. [\[CrossRef\]](#)

49. Iqbal, M.; Azam, M.; Naeem, M.; Khwaja, A.S.; Anpalagan, A. Optimization classification, algorithms and tools for renewable energy: A review. *Renew. Sustain. Energy Rev.* **2014**, *39*, 640–654. [[CrossRef](#)]
50. Sinha, S.; Chandel, S.S. Review of recent trends in optimization techniques for solar photovoltaic-wind based hybrid energy systems. *Renew. Sustain. Energy Rev.* **2015**, *50*, 755–769. [[CrossRef](#)]
51. Lande-Sudall, D.; Stallard, T.; Stansby, P. Co-located deployment of offshore wind turbines with tidal stream turbine arrays for improved cost of electricity generation. *Renew. Sustain. Energy Rev.* **2019**, *104*, 492–503. [[CrossRef](#)]
52. Oloo, F.; Molefyan, B.; Rampokanyo, M. Modelling and optimisation studies for generator dispatch strategies for deployment of an off-grid micro-grid in South Africa. In Proceedings of the 2020 6th IEEE International Energy Conference, Gammarth, Tunisia, 28 September–1 October 2020; pp. 533–538. [[CrossRef](#)]
53. Prinsloo, G.; Dobson, R.; Brent, A. Scoping exercise to determine load profile archetype reference shapes for solar co-generation models in isolated off-grid rural African villages. *J. Energy S. Afr.* **2016**, *27*, 11–27. [[CrossRef](#)]
54. Rautenbach, C.; Barnes, M.A.; de Vos, M. Tidal characteristics of South Africa. *Deep. Res. Part I Oceanogr. Res. Pap.* **2019**, *150*, 103079. [[CrossRef](#)]
55. Mtukushe, N.F.; Ojo, E.E. The Study of Electrical Power Generation from Tidal Energy in South Africa. In Proceedings of the 2021 Southern African Universities Power Engineering Conference/Robotics and Mechatronics/Pattern Recognition Association of South Africa (SAUPEC/RobMech/PRASA), Potchefstroom, South Africa, 27–29 January 2021; pp. 7–12. [[CrossRef](#)]
56. Rehman, S.; Natarajan, N.; Mohandes, M.A.; Meyer, J.P.; Alam, M.M.; Alhems, L.M. Wind and wind power characteristics of the eastern and southern coastal and northern inland regions, South Africa. *Environ. Sci. Pollut. Res.* **2022**, *29*, 85842–85854. [[CrossRef](#)] [[PubMed](#)]
57. Creamer, T. Global Wind Body Urges South Africa to Explore ‘Excellent’ Offshore Wind Potential. *Page Magazine*, 10 January 2024.
58. Hu, E.; Gao, S.; Jia, X.; Liu, L.; Ren, A.; Li, Q.; Wang, S.; Shen, Z.; Xing, Y. Research on capacity optimization of offshore wind power flow combined power generation system based on homer. In Proceedings of the 2020 IEEE 4th Conference on Energy Internet and Energy System Integration (EI2), Wuhan, China, 30 October–1 November 2020; pp. 2684–2689. [[CrossRef](#)]
59. Toumi, S.; Amirat, Y.; Elbouchikhi, E.; Zhou, Z.; Benbouzid, M. Techno-Economic Optimal Sizing Design for a Tidal Stream Turbine—Battery System. *J. Mar. Sci. Eng.* **2023**, *11*, 679. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.